

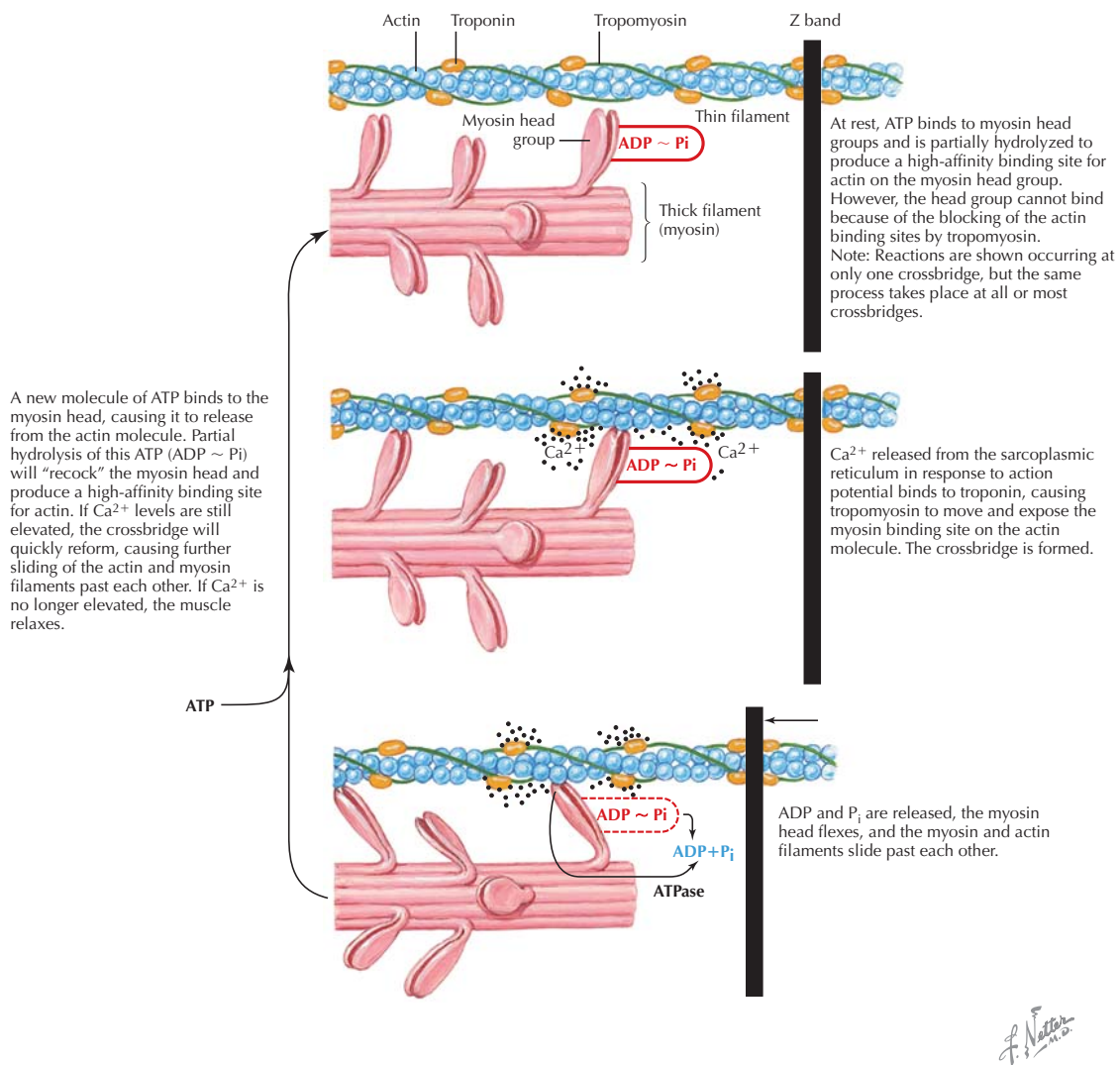


Figure 3.15 Biochemical Mechanics of Muscle Contraction Muscle contraction is produced when actin and myosin form and recycle crossbridges. The process is dependent on the presence of free intracellular Ca²⁺ and the availability of ATP.

Ca²⁺ concentration. Some of these substances produce depolarization of the cell membrane, resulting in opening of voltage-gated membrane Ca²⁺ channels and release of Ca²⁺ from intracellular stores, in a process similar to that in skeletal muscle. In **pharmacomechanical coupling**, binding of a ligand to a membrane receptor produces an increase in intracellular Ca²⁺ concentration, and thus smooth muscle contraction, without altering membrane potential. An example is contraction of vascular smooth muscle by binding of norepinephrine or epinephrine to α -adrenergic receptors.

Depolarization or pharmacomechanical coupling leads to elevation of intracellular Ca²⁺, the common signal in smooth muscle contraction (Fig. 3.18). In the latter case, ligand binding activates membrane phospholipase C, which cleaves



In **phasic smooth muscle**, contractions are short in duration, as in skeletal muscle. In **tonic smooth muscle**, tension may be maintained for long periods of time with little ATP utilization. In this case, myosin is dephosphorylated while attached to actin, forming the "latch state" (see Fig. 3.18). In this state, crossbridge recycling is very slow and ATP utilization is reduced. This latch state is important in maintaining tension in blood vessels and sphincters.

inositol biphosphate, thereby producing IP₃, which releases Ca²⁺ from intracellular stores. Whether elevated by membrane depolarization or activation of the inositol pathway, Ca²⁺ binds to the protein **calmodulin**, and the Ca²⁺-calmodulin

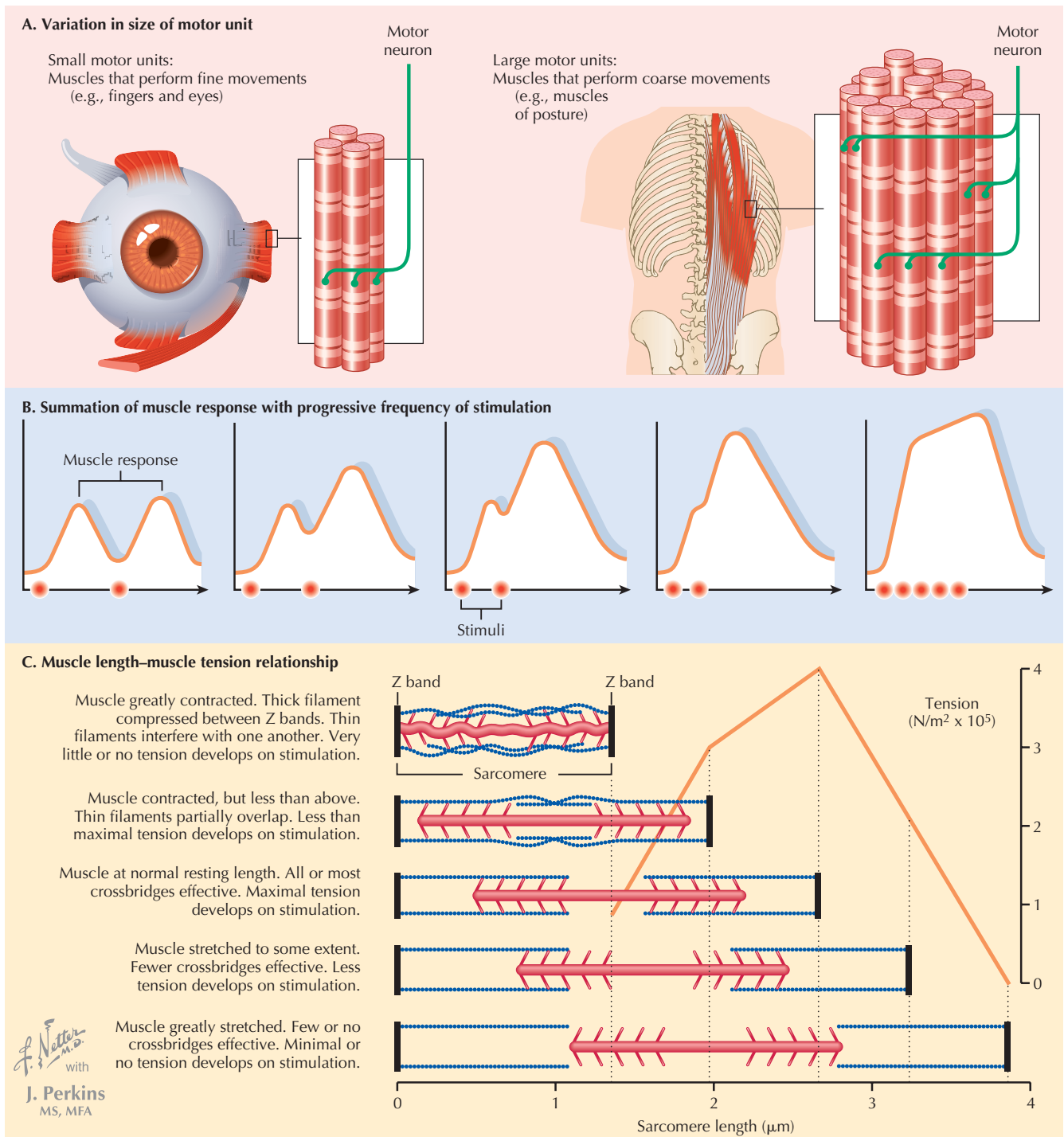


Figure 3.16 Grading of Muscle Tension and Length-Tension Relationship The force generated when skeletal muscle is stimulated is related to the size of the motor units stimulated (A), the number of motor units activated and the frequency of stimulation of the muscle fibers (B), and the resting length of muscle fibers (C). Small motor units are involved in fine motor control, for example in fingers and eyes, while larger motor units are involved in coarser movements (A). Skeletal muscles also produce more force when more motor units are recruited. With repeated stimulation of a muscle fiber, summation occurs, in which individual twitches occur without complete relaxation between twitches (B). There is also a length-tension relationship, whereby greater resting sarcomere length (stretch of the muscle before contraction) is associated with greater force of contraction, up to an optimal resting length (C).

Table 3.3 Comparison of Muscle Structure and Function

	Skeletal Muscle	Cardiac Muscle	Smooth Muscle
STRUCTURE			
Morphology	Long; cylindrical	Branched	Spindle or fusiform
Nuclei	Multiple; located peripherally	One (sometimes two); located centrally	One; located centrally
Sarcomere	Yes; striated pattern	Yes; striated pattern	No
T tubules	Yes; forms triad with sarcoplasmic reticulum	Yes; forms dyad with sarcoplasmic reticulum	No; caveolae
Electrical coupling of cells	No	Yes; intercalated discs contain gap junctions	Yes; gap junctions
Regeneration	Yes; via satellite cells	No	Yes
Mitosis	No	No	Yes
PHYSIOLOGY			
Extracellular Ca^{2+} required for contraction	No	Yes	Yes
Regulation of crossbridge formation	Ca^{2+} binding to troponin	Ca^{2+} binding to troponin	Ca^{2+} -calmodulin activation of myosin kinase and phosphorylation of myosin
Control of contraction	Motor neurons	Autonomic nerves; β -adrenergic agonists	Autonomic nerves; hormones
Summation of twitches by increased stimulus frequency	Yes	No*	Yes
Tension varies with filament overlap	Yes	Yes	Yes

Major differences in structure and function of skeletal, cardiac, and smooth muscle are indicated.

*Cardiac muscle cannot be tetanized, but the force of contraction will increase at high stimulus frequency because of an increase in intracellular $[\text{Ca}^{2+}]$, a phenomenon termed "Treppe."

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complex activates the enzyme **myosin kinase**, allowing interaction between myosin and actin, producing contraction as these proteins slide past each other. The contraction cycle continues as long as intracellular Ca^{2+} is elevated.

Relaxation of Smooth Muscle

Smooth muscle *relaxation* may also be induced by pharmacomechanical coupling, in which case various substances stimulate a rise in intracellular cAMP or cGMP. Ultimately, intracellular Ca^{2+} is reduced, causing relaxation. An example is relaxation produced by binding of epinephrine or norepinephrine to β -adrenergic receptors on vascular smooth muscle.

CARDIAC MUSCLE

Cardiac muscle is similar to skeletal muscle or smooth muscle in some respects, and dissimilar in others (Fig. 3.19 and Table 3.3). Contraction of skeletal muscle is under the voluntary control of the central nervous system, whereas contraction of cardiac and smooth muscle is involuntary. Both unitary smooth muscle and cardiac muscle have the capability for spontaneous electrical activity; cardiac contraction is normally under control of cardiac pacemaker cells in the sinoatrial (SA) node. Gap junctions in cardiac muscle, as in unitary smooth muscle, allow synchronous contraction. These gap junctions in cardiac muscle are found in the **intercalated disks** between cells. Cardiac and skeletal muscles have highly organized sarcomeres, leading to a striated appearance.

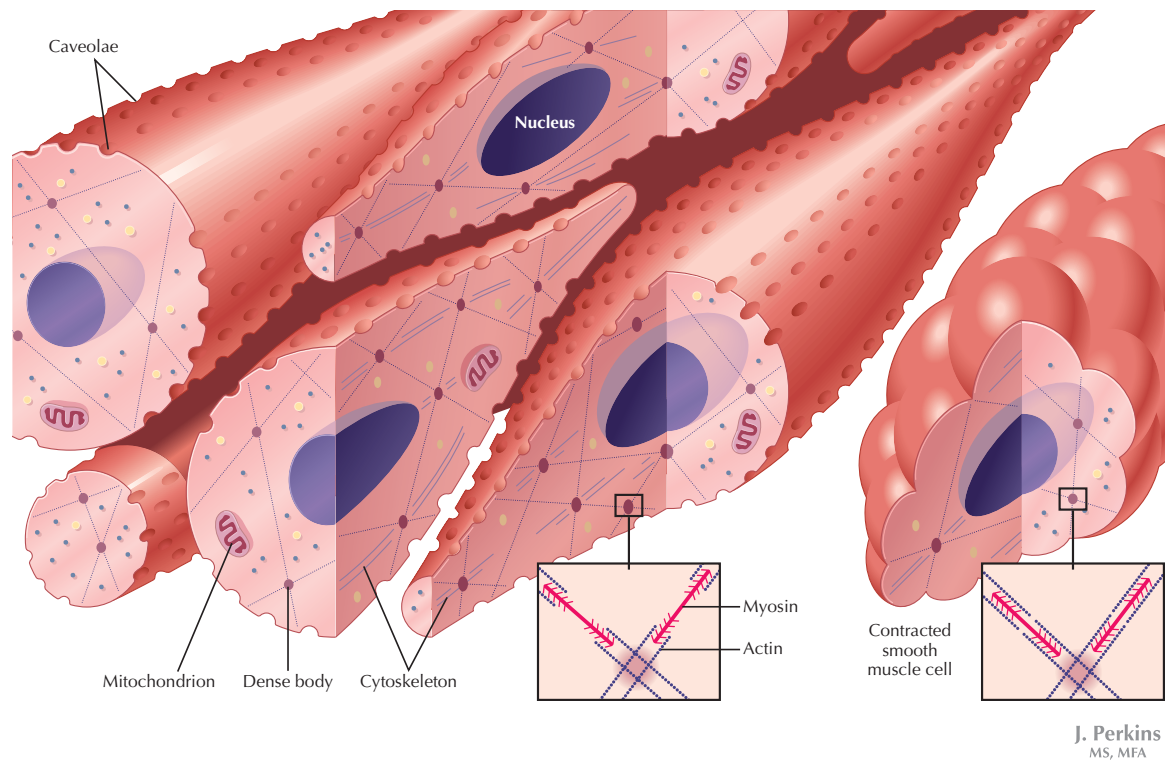


Figure 3.17 Smooth Muscle Structure Actin and myosin filaments of spindle-shaped smooth muscle cells are arranged quite differently than contractile proteins in skeletal muscle. Actin is anchored to dense bodies within the muscle cell and to the plasma membrane, and the sliding of myosin and actin filaments produces contraction when intracellular Ca^{2+} is elevated, either by release from intracellular stores or flux through Ca^{2+} channels. Caveolae are invaginations in the cell membrane and are a site of Ca^{2+} flux.

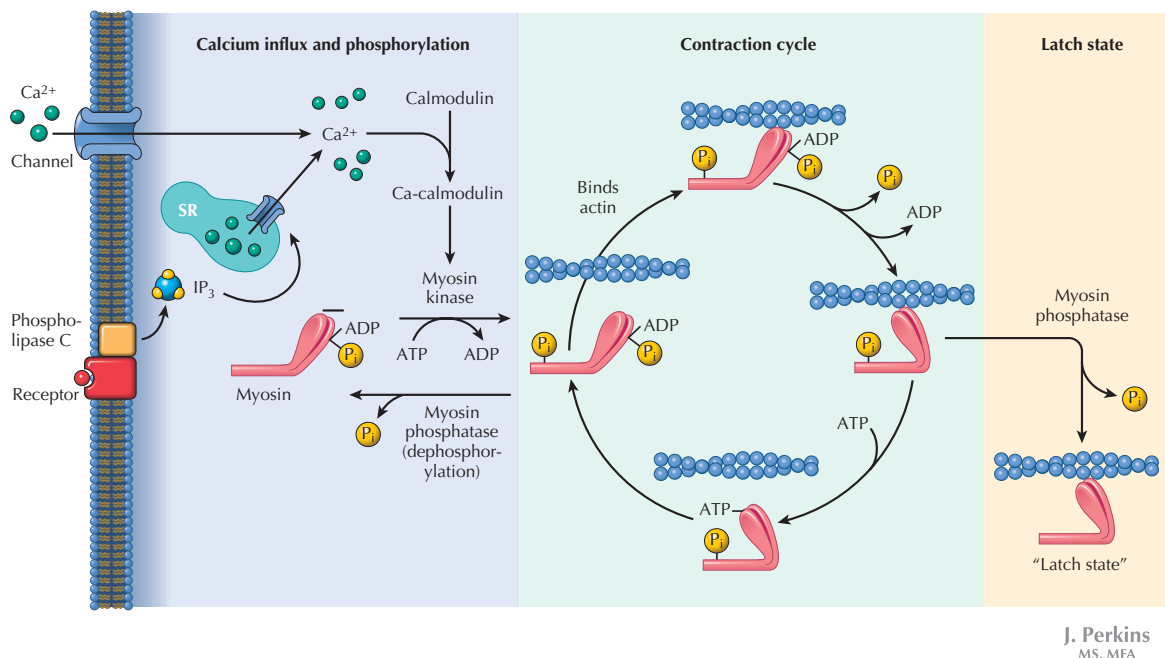


Figure 3.18 Excitation-Contraction Coupling of Smooth Muscle Binding of a ligand to the sarcolemma results in elevation of free intracellular Ca^{2+} , through either depolarization of the cell membrane and opening of Ca^{2+} channels, or activation of the enzyme phospholipase C. In the latter case, cleavage of inositol trisphosphate by phospholipase C produces IP_3 , which binds to sites on the sarcoplasmic reticulum, causing release of stored Ca^{2+} . In either case, Ca^{2+} binds to the protein calmodulin, which activates myosin kinase, initiating actin-myosin interaction. The contraction cycle continues as long as Ca^{2+} is elevated. The latch state occurs when myosin is dephosphorylated by myosin phosphatase. In this state, contraction can be maintained without further ATP hydrolysis.

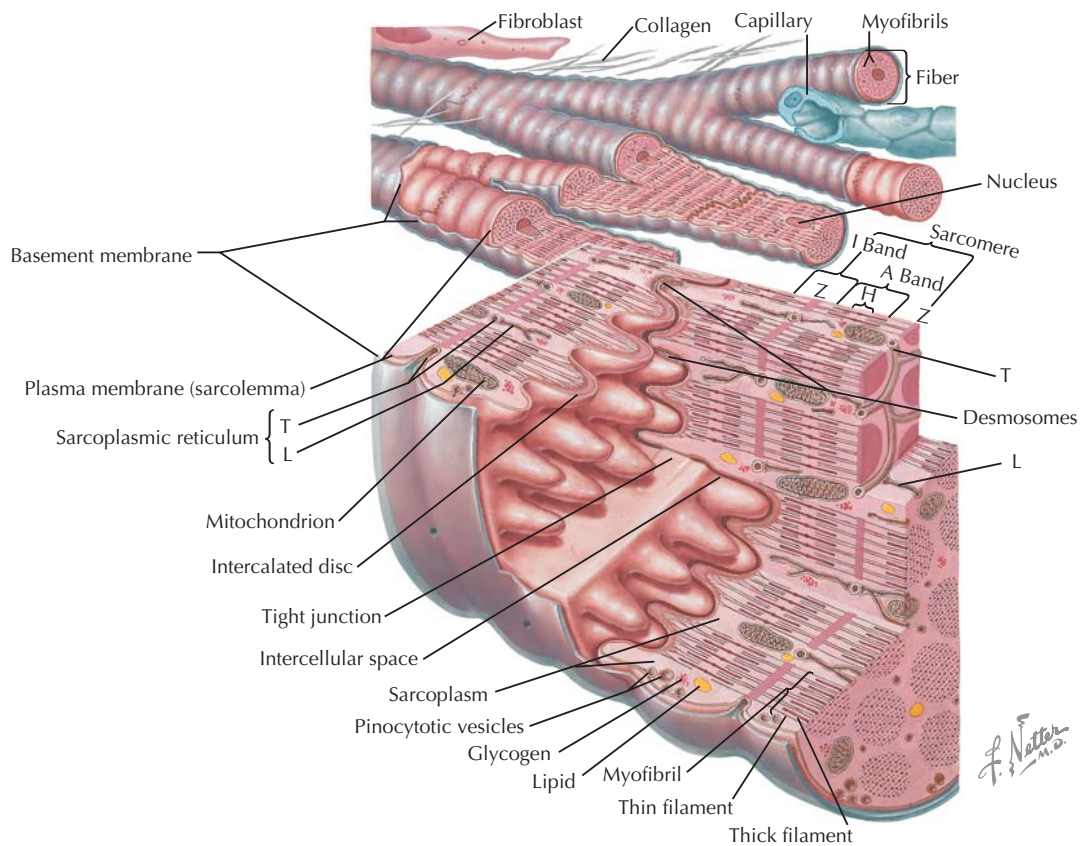


Figure 3.19 Schema of Structure of Cardiac Muscle The striated appearance of cardiac muscle is associated with the arrangement of the contractile proteins actin and myosin into sarcomeres, as in skeletal muscle. A notable difference between these types of muscles is that cardiac muscle utilizes extracellular Ca^{2+} as well as intracellular stores to initiate contraction. Another difference is that transverse tubules of cardiac muscle form dyads with the terminal cisternae of the sarcoplasmic reticulum, as opposed to the triads found in skeletal muscle. In addition, unlike skeletal muscle, cardiac muscle fibers are connected by gap junctions at intercalated disks between the cells, allowing the spread of depolarization from cell to cell, producing synchronous contraction of muscle.



Calcium channel blockers are drugs that block voltage-dependent Ca^{2+} channels (L-type Ca^{2+} channels). These drugs are often used in antihypertensive therapy, based on their effects on vascular smooth muscle. However, because they also block voltage-dependent Ca^{2+} channels of the heart, they slow conduction in the heart and reduce myocardial contractility. Calcium channel blockers may belong to several classes of chemicals, including dihydropyridines. Nifedipine and amlodipine are examples of the dihydropyridine class. Voltage-gated Ca^{2+} channels of T-tubules in cardiac and skeletal muscle are also referred to as dihydropyridine receptors, based on their ability to bind these drugs. Note, however, that while cardiac muscle contractility is reduced by calcium blockers, skeletal muscle contractility is unaffected. Skeletal muscle is not dependent on extracellular Ca^{2+} for contraction.

The velocity of contraction of skeletal muscle is dependent on muscle fiber type (fast twitch vs. slow twitch fibers); cardiac muscle contraction is slower than that of skeletal muscle, but more rapid than contraction in smooth muscle. Cardiac and smooth muscle utilize both intracellular and extracellular sources of Ca^{2+} , whereas the only Ca^{2+} source for contraction of skeletal muscle is intracellular (from the sarcoplasmic reticulum). As in skeletal muscle, Ca^{2+} in cardiac muscle binds to troponin to initiate crossbridge formation.

Other aspects of cardiac muscle function, including mechanical function and regulation of pacemaker activity, are considered in Section 3.

CHAPTER 4

Organization and General Functions of the Nervous System

The nervous system consists of the **central nervous system (CNS)** and the **peripheral nervous system (PNS)**. The CNS includes the brain and spinal cord. The peripheral nervous system includes nerves, ganglia, and sensory receptors outside the CNS. The peripheral nervous system can also be subdivided into **sensory** and **motor** divisions. Sensory nerves transmit information from various sensory receptors to the CNS; motor nerves transmit information from the CNS to muscles and glands, thereby controlling their activity.

THE CENTRAL NERVOUS SYSTEM

The general structure of the central nervous system and brain is illustrated in Figure 4.1.

Brain

The brain can be subdivided into the **telencephalon** (also known as the cerebrum or cerebral hemispheres), **diencephalon** (thalamus, hypothalamus), **cerebellum**, and **brainstem** (midbrain, pons, and medulla).

Blood-Brain Barrier

The environment of neurons within the CNS is maintained in part by the **blood-brain barrier**. Endothelial cells of capillaries within the CNS are joined by tight junctions, preventing the movement of water-soluble substances, highly charged molecules, and cells between blood and brain. Astrocytes (non-neuronal cells of the CNS) are also involved in maintaining the integrity of the blood-brain barrier.

Cerebrospinal Fluid

The formation, circulation, and regulation of the composition of **cerebrospinal fluid (CSF)** compose a second important factor in the homeostasis of the environment within the CNS. CSF, which is secreted by ependymal (epithelial) cells of the choroid plexus, differs from blood plasma somewhat in composition and circulates through the ventricles of the brain and the subarachnoid space around the brain and spinal cord (Table 4.1 and Figs. 4.2 and 4.3). Specifically, the CSF has a lower concentration of bicarbonate anion than plasma, result-

ing in lower pH. This difference makes the CSF sensitive to changes in PCO_2 of the blood, an important factor in regulation of respiration by the brain. Substances within the CSF are freely exchanged with the interstitial fluid of the brain, in contrast to substances within blood plasma. Because CSF is secreted at a rate of approximately 500 mL/day and the total volume of CSF is approximately 150 mL, it turns over at a rate of three to four times per day.

Telencephalon

In the telencephalon, the right and left cerebral hemispheres consist of the outer cerebral cortex (gray matter) and inner white matter. (Gray matter contains unmyelinated axons, and white matter contains myelinated axons). The hemispheres are linked anatomically and functionally by bundles of nerves known as commissures: the large corpus callosum and smaller **anterior**, **posterior**, and **hippocampal commissures** (Fig. 4.4).

The cerebral cortex consists of five major areas, based on gross anatomical features: the frontal, parietal, temporal, and occipital lobes and the insula (Fig. 4.5). It has afferent and efferent connections to the thalamus and basal ganglia (two sets of nuclei deep in the cerebral hemispheres, connecting with the cerebrum, thalamus, and brainstem) and many other regions of the brain. Regions within the cerebral cortex receive and integrate sensory information, integrate motor function, and perform other high-level functions such as learning and reasoning (see Fig. 4.4). Much of the sensory information is received indirectly, passing through the thalamus, except in



The term **neocortex** refers to the outermost layer of the cerebral hemispheres. This area is the most recently evolved part of the cerebral cortex and is only a few millimeters thick. The neocortex is considered the highest center in the brain, where the most complex functions ascribed to the cerebral cortex are performed, including sensory perception, initiation of motor function, language skills, and conscious thought.

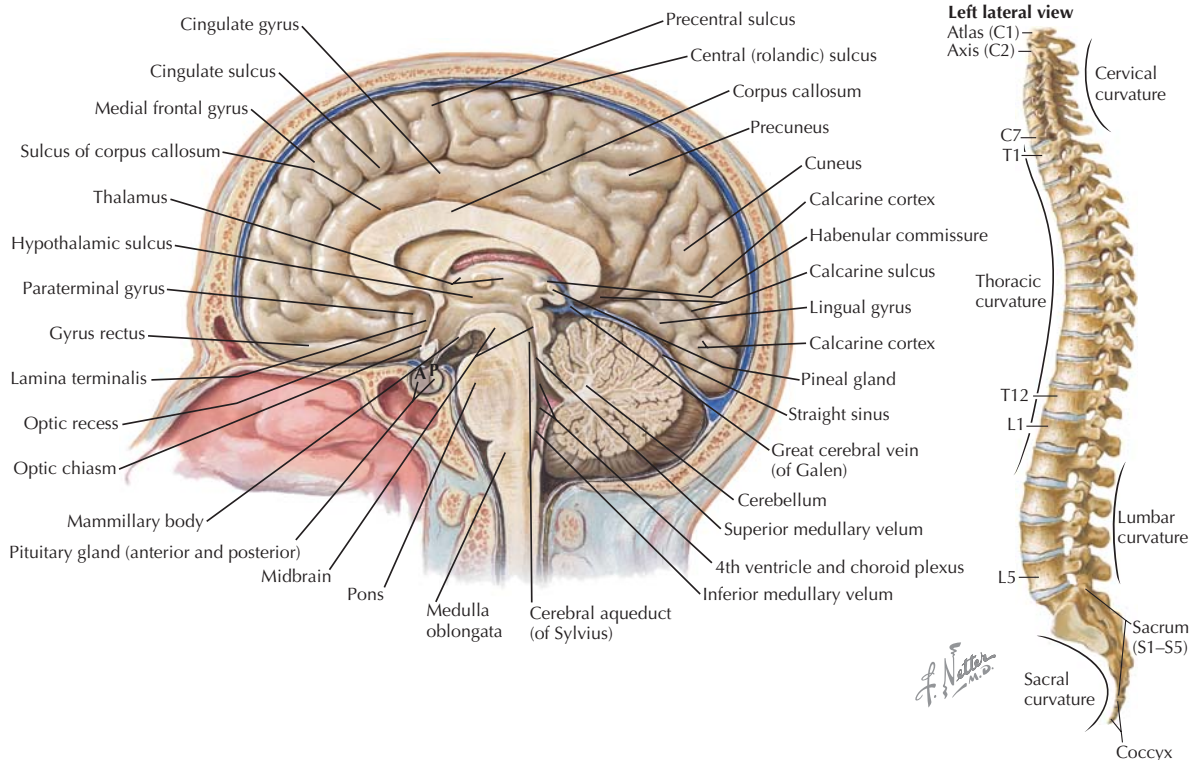


Figure 4.1 Major Parts of the Central Nervous System and the Vertebral Column The brain is illustrated as a midsagittal section, with the frontal, parietal, and occipital lobes of the cerebral cortex (outer layer of the cerebral hemispheres) labeled (the temporal lobes are not seen in this section), as well as other major divisions of the brain. The vertebral column consists of cervical, thoracic, lumbar, and sacral vertebrae. It houses the spinal cord, which extends from the medulla oblongata to lumbar level L1–L2.

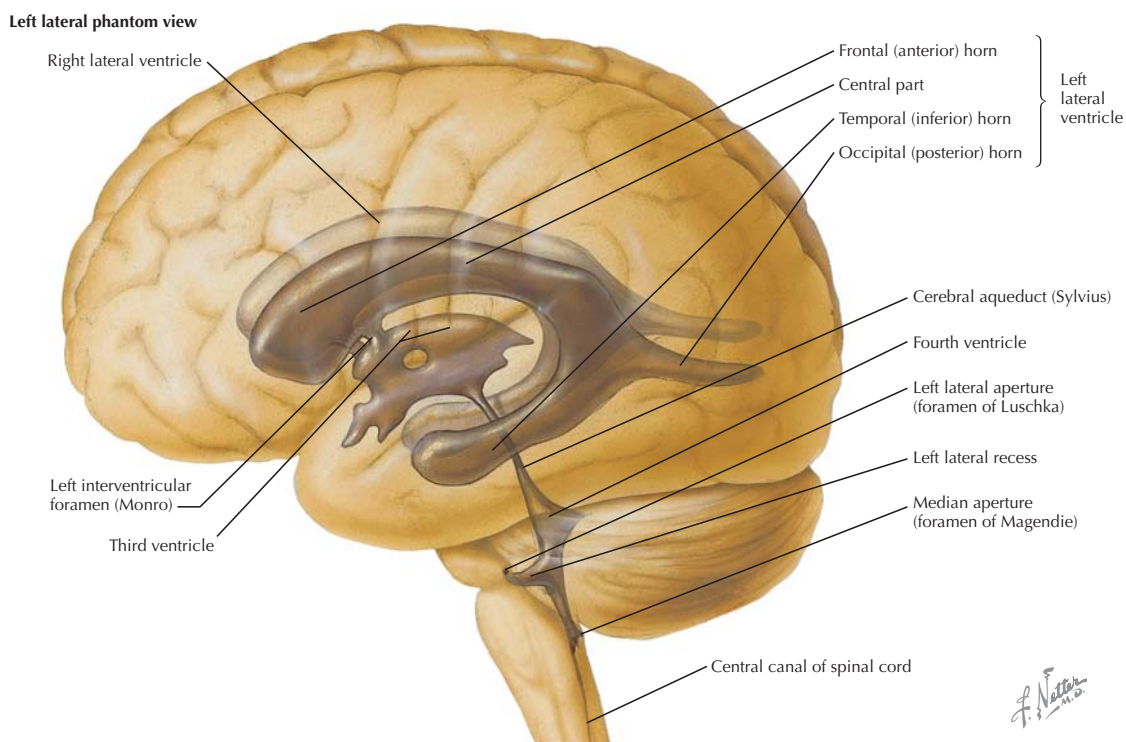


Figure 4.2 Brain Ventricles and CSF Composition Circulation of CSF through the four ventricles of the brain and in the subarachnoid space is essential for maintaining the environment of the brain and spinal cord. The composition of CSF, which is secreted by the choroid plexus, differs from that of plasma (see Table 4.1).

the case of olfactory signals. Generally, the right and left cerebral hemispheres receive input from the contralateral side of the body.

The basal ganglia consist of two sets of nuclei (in the right and left hemispheres) located deep within the cerebral hemispheres. With connections to the various lobes of the cerebral cortex, thalamus, and brainstem, the basal ganglia are involved in regulation of movement, among other functions. Two additional deep formations, the **hippocampus** and **amygdala**, are located within the medial temporal lobes. Both are part of the broader limbic system, involved in emotion and long-term memory and affecting the endocrine and autonomic nervous systems (Fig. 4.6). The hippocampus has an important role in memory and spatial navigation; the amygdala is involved in emotion. Through its connections to the hypothalamus, the hippocampus also affects sympathetic nervous system function.

Table 4.1 Comparison of CSF and Blood Composition

Parameter	CSF	Blood Plasma
Na ⁺ (mEq/L)	140-145	135-147
K ⁺ (mEq/L)	3	3.5-5.0
Cl ⁻ (mEq/L)	115-120	95-105
HCO ₃ ⁻ (mEq/L)	20	22-28
Glucose (mg/dL)	50-75	70-110
Protein (g/dL)	0.05-0.07	6.0-7.8
pH	7.3	7.35-7.45

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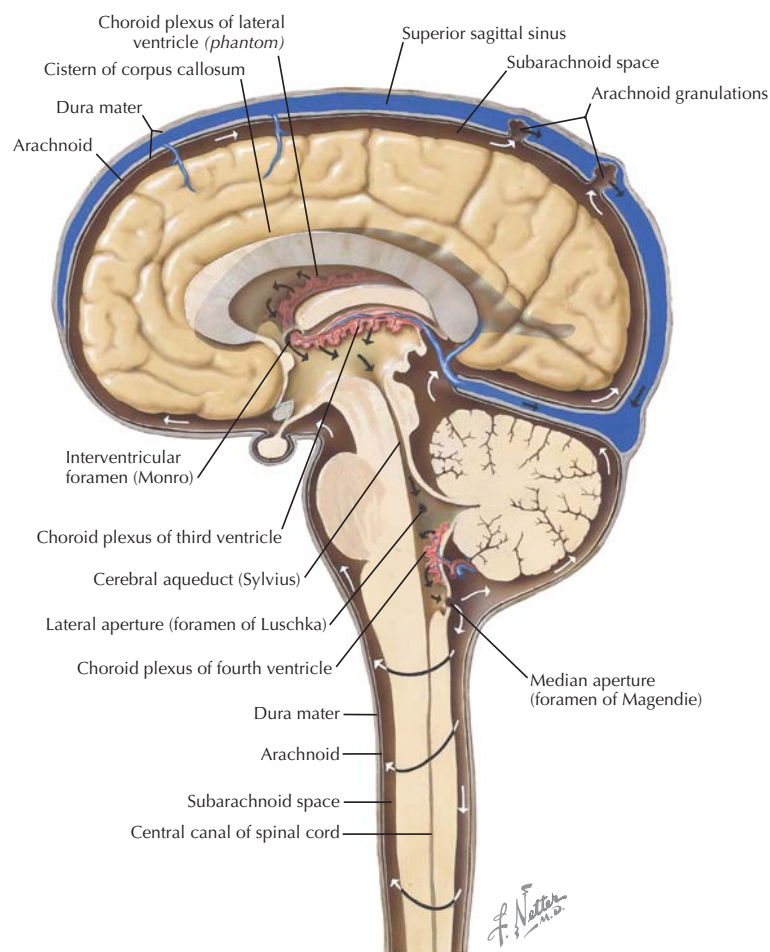


Figure 4.3 Circulation of CSF CSF, produced by the choroid plexus, circulates through the two lateral ventricles and third and fourth ventricles, leaving the fourth ventricle through the lateral and medial apertures and entering the subarachnoid space. Much of the fluid is reabsorbed at the arachnoid granulations, into the venous system, as well as into the capillaries of the CNS and pia mater.

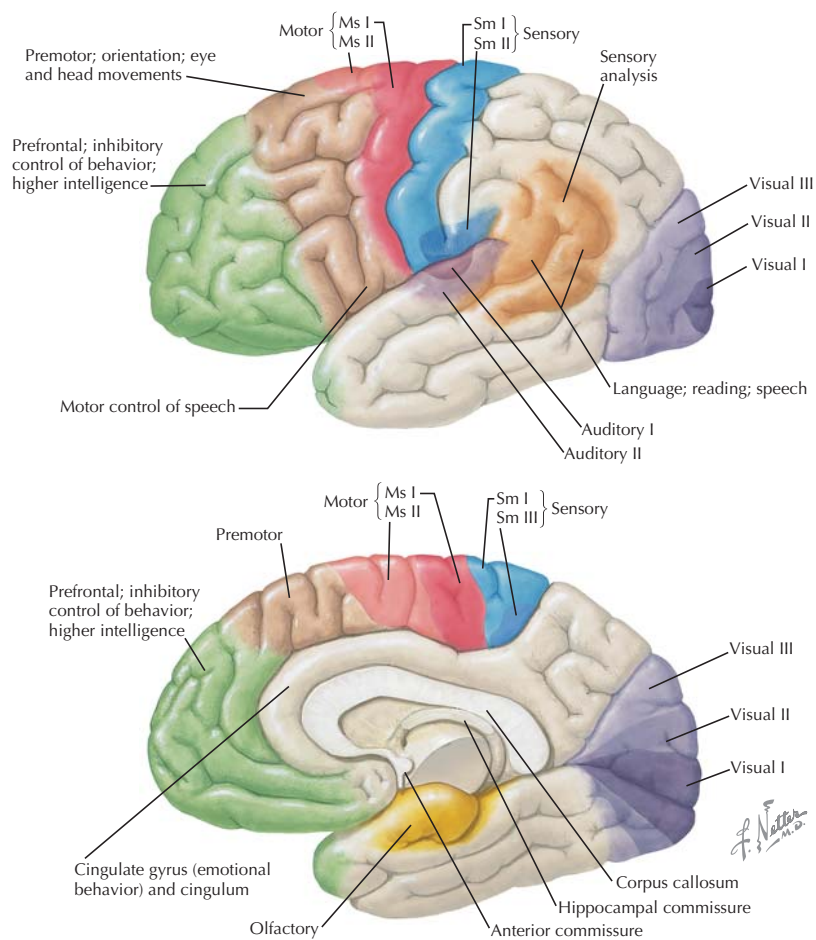


Figure 4.4 Cerebral Cortex: Association Pathways and Localization of Function The two cerebral hemispheres are linked functionally and anatomically by the corpus callosum and anterior, posterior, and hippocampal commissures. In the cerebral cortex, functional regions involved in sensory and motor activity as well as integration of information from various sources have been identified. The cortex is essential for high-level intellectual functions including learning, reasoning, memory and language skills, and conscious thought and is involved in planning and execution of fine motor activity.

Table 4.2 Major Functions of the Hypothalamus

Hypothalamic Area	Major Functions*
Preoptic and anterior	Heat loss center: cutaneous vasodilation and sweating
Posterior	Heat conservation center: cutaneous vasoconstriction and shivering
Lateral	Feeding center: eating behavior
Ventromedial	Satiety center: inhibits eating behavior
Supraoptic (subfornical organ and organum vasculosum)	ADH and oxytocin secretion (sensation of thirst)
Paraventricular	ADH and oxytocin secretion
Periventricular	Releasing hormones for the anterior pituitary

*Stimulation of the center causes the responses listed.
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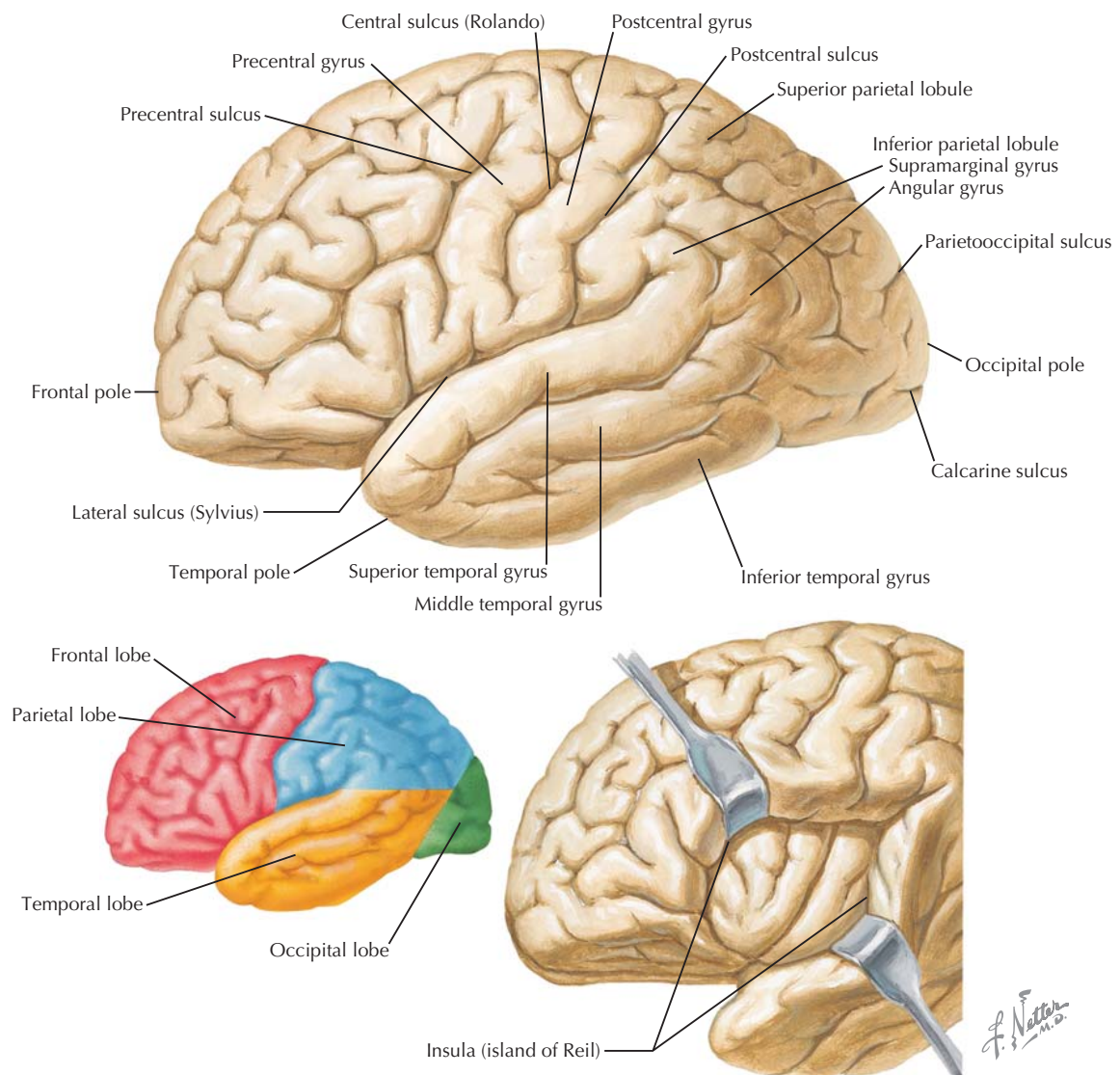


Figure 4.5 Organization of the Brain: Surface Anatomy of the Cerebral Cortex The outer layers of the cerebrum (telencephalon) are the gray matter, containing neurons with unmyelinated fibers. The cerebral cortex is only 2 to 4 mm thick, but is highly convoluted in primates, such that most of the cortex is actually buried in sulci (grooves). Four lobes are discernable from the surface; the fifth lobe, the insula, can be viewed by retracting the outer cortex at the lateral fissure. The cerebral cortex is the highest center of the brain and is involved in learning, memory, and reasoning, as well as motor and sensory control and integration.

Diencephalon

The **diencephalon**, consisting of the thalamus and hypothalamus, is located between the cerebral hemispheres and the brainstem and is part of the limbic system (see Fig. 4.6). The thalamus processes sensory input and relays it to the cerebral cortex and processes motor signals leaving the cerebral cortex. The hypothalamus has a central role in the regulation of body temperature, the reproductive system, hunger and thirst, salt and water balance, circadian rhythms,

the autonomic nervous system, and endocrine function (Table 4.2).

Cerebellum

The **cerebellum** is located between the cerebral cortex and the spinal cord and in close proximity to the brainstem (see Fig. 4.1). It integrates sensory and motor information as well as information from the inner ear and has an important role in **proprioception** (unconscious perception of posture, spatial

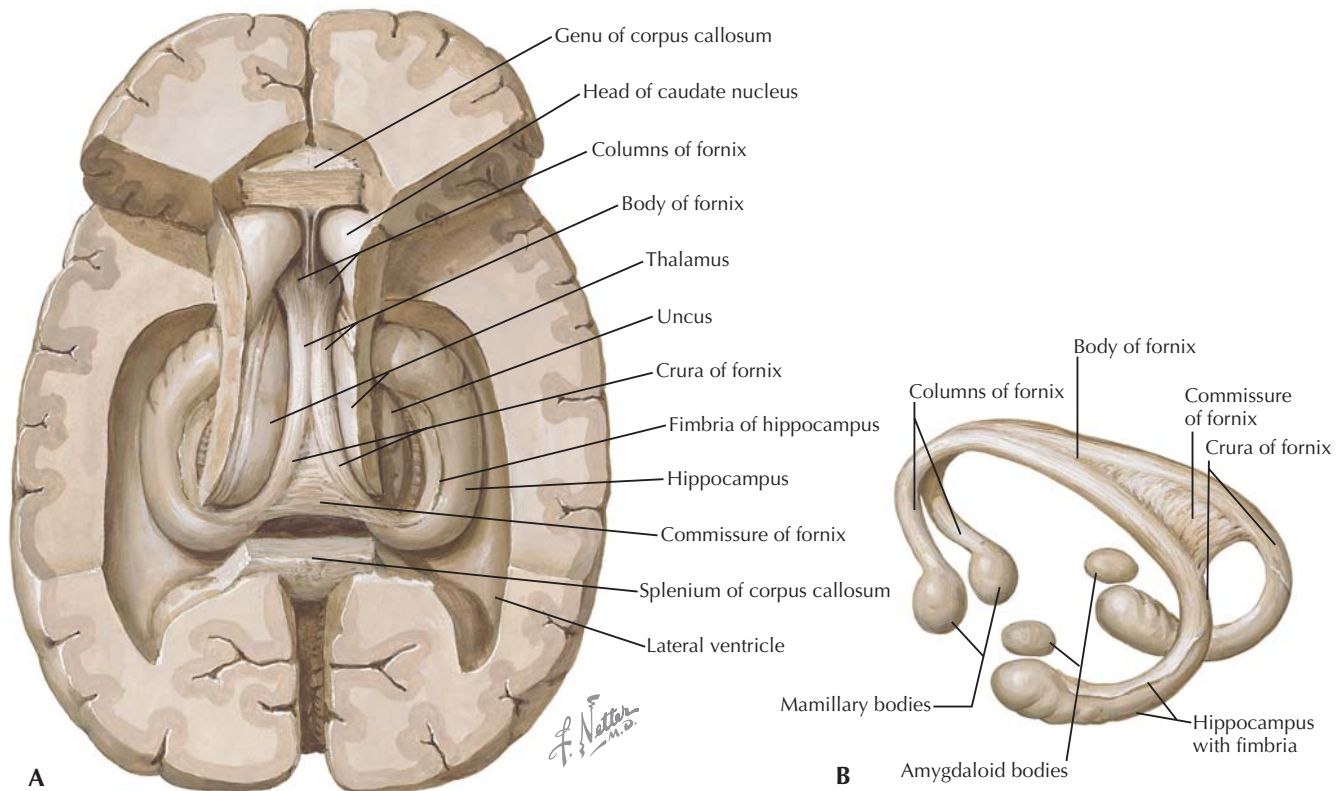


Figure 4.6 The Limbic System The limbic system consists of the hypothalamus and several structures forming a border or ring (limbus) around the diencephalon. It includes the cingulate, parahippocampal, and subcallosal gyri, and the amygdala, hypothalamus, hippocampi, and mamillary bodies. *A*, Cerebral cortical and white matter have been removed, as well as the corpus callosum and part of the thalamus and caudate nucleus. *B*, The fornix consists of curved bundles of fibers connecting the mamillary bodies and hippocampi. The limbic system has an important role in emotion and the linking of emotion with memory, as well as endocrine and autonomic function.

orientation, and movement). Proprioception is based on input from proprioceptors within muscles, joints, tendons, and the inner ear.

Brainstem

The **brainstem** is the lowest portion of the brain, consisting of the midbrain, pons, and medulla oblongata (medulla). The medulla, which is anatomically continuous with the spinal cord, contains centers that regulate autonomic functions. This includes centers involved in the regulation and integration of cardiovascular and respiratory functions, and swallowing, vomiting, and coughing reflexes. The pons, situated above (rostral to) the medulla, is also involved in regulation of breathing, and relays sensory information from the cerebrum to the cerebellum. The midbrain, also known as the **mesencephalon**, is the most rostral portion of the brainstem. It is involved in eye movement and transmission of visual and auditory information. It is also the location of the **substantia nigra**, which, in association with the basal ganglia, has a role

in regulation of motor activity. **Cranial nerves III–XII (CN III–XII)** originate from the brainstem.

The Spinal Cord

The spinal cord originates at the medulla at the base of the skull and extends within the vertebral column from the medulla oblongata to the lumbar region (Fig. 4.7). It consists of neural tissues covered by three meninges (membranes) that are continuous with those of the brain:

- the inner pia mater
- the middle arachnoid membrane
- the outer dura mater

Cerebrospinal fluid is found between the arachnoid membrane and pia mater. The spinal cord gives rise to 31 pairs of spinal nerves, organized into three plexuses:

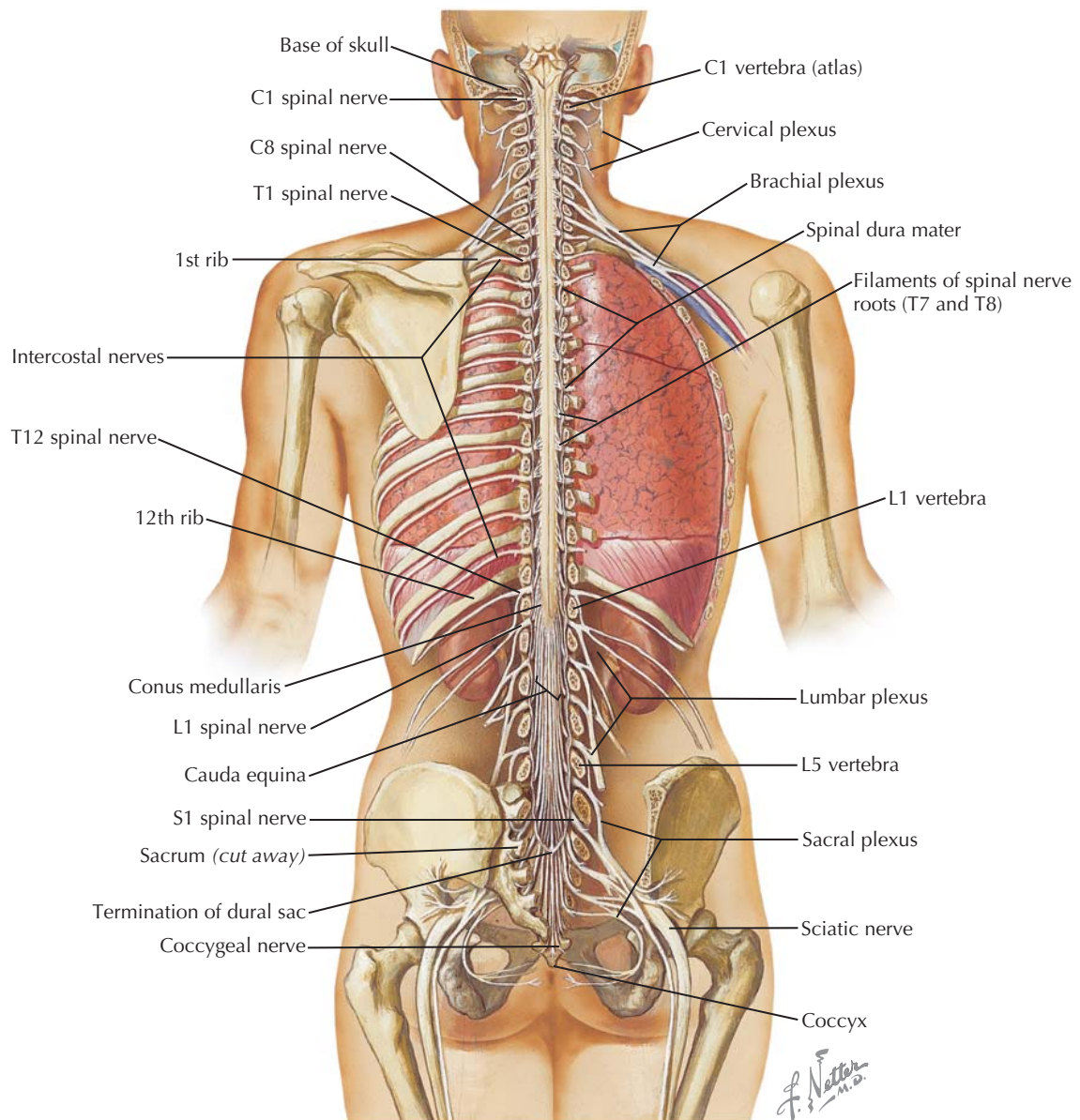


Figure 4.7 Spinal Cord and Ventral Rami In Situ The 31 pairs of spinal nerves originating from the spinal cord are organized into the cervical, brachial, and lumbosacral plexuses. Fibers from the plexuses are distributed, respectively, to the neck, upper limbs, and pelvis and lower limbs. Efferent fibers innervate skeletal muscle, whereas afferent fibers carry sensory information from the skin, muscle, and joints to the CNS.

- The **cervical plexus**, which forms nerves distributed to the back of the head and the neck.
- The **brachial plexus**, which forms nerves distributed to the upper limbs.
- The **lumbosacral plexus**, which forms nerves distributed to the pelvis and lower limbs.

The anatomy of spinal membranes and nerve roots is illustrated in Figure 4.8.

THE PERIPHERAL NERVOUS SYSTEM

The peripheral nervous system is the portion of the nervous system that lies beyond the brain and spinal cord. It consists of:

- Motor neurons that convey impulses from the CNS to effector tissues or organs.
- Sensory neurons that convey impulses from peripheral sensory receptors to the CNS.

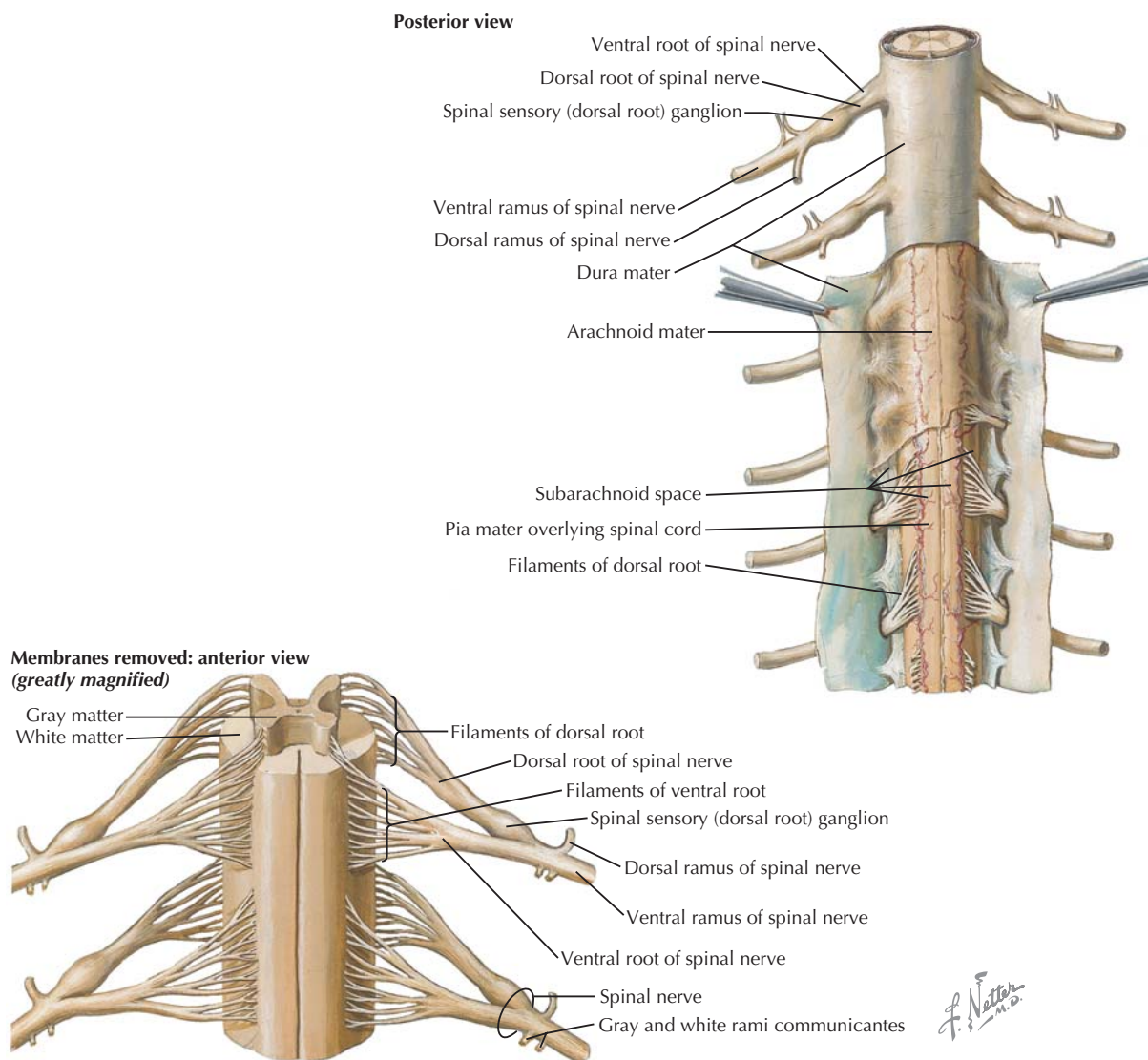


Figure 4.8 Spinal Membranes and Nerve Roots Motor fibers and sensory fibers pass through 31 pairs of spinal nerves. The outer dura mater, arachnoid mater, and inner pia mater are the three coverings of the spinal cord, with CSF circulating in the subarachnoid space.

The peripheral nervous system is subdivided into somatic and autonomic components. In the somatic portion of the peripheral nervous system, motor fibers of the spinal nerves innervate skeletal muscles; sensory information from receptors in the skeletal muscles, joints, and skin is transmitted through sensory fibers to the CNS (Fig. 4.9). The autonomic

component of the peripheral nervous system is composed of motor and sensory fibers involved in the involuntary control of homeostatic mechanisms through actions on visceral organs. The autonomic nervous system and its sympathetic and parasympathetic divisions are discussed further in Chapter 7.

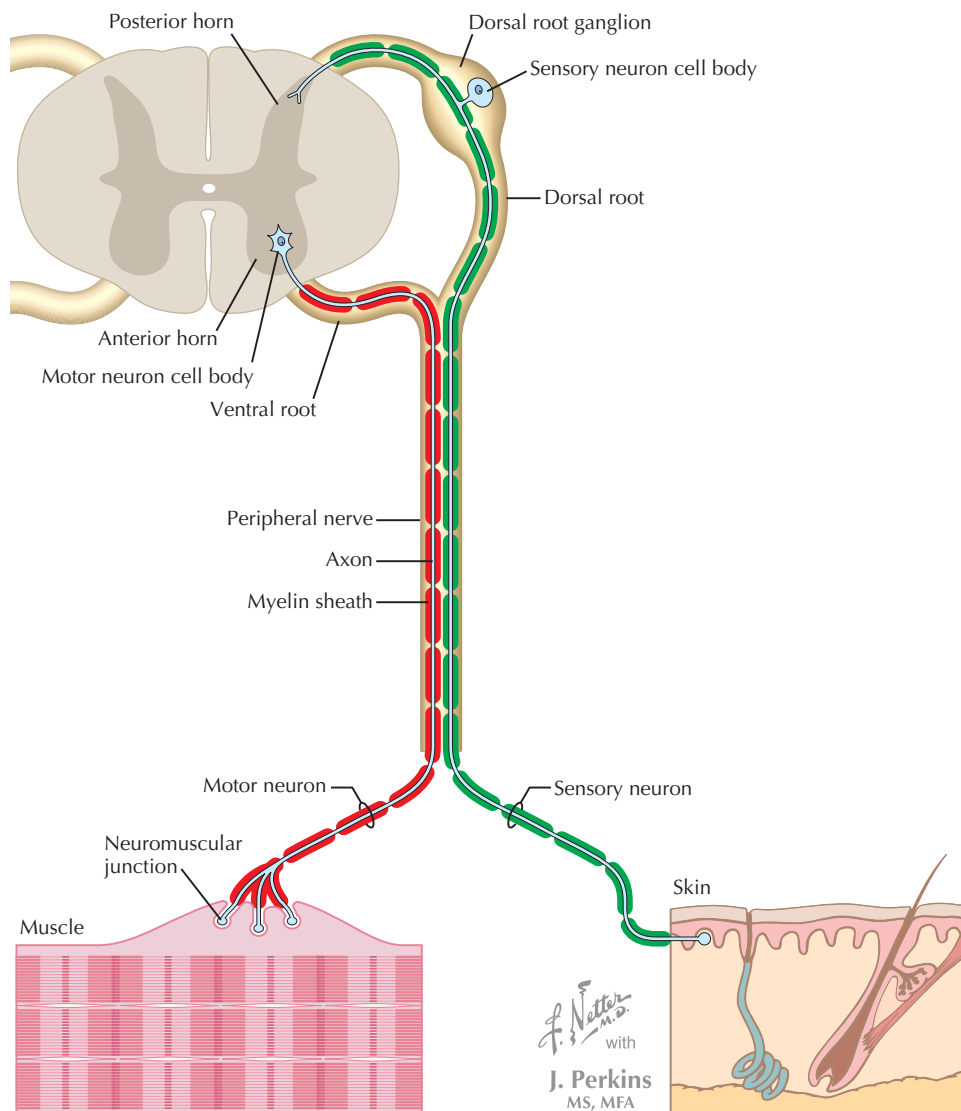


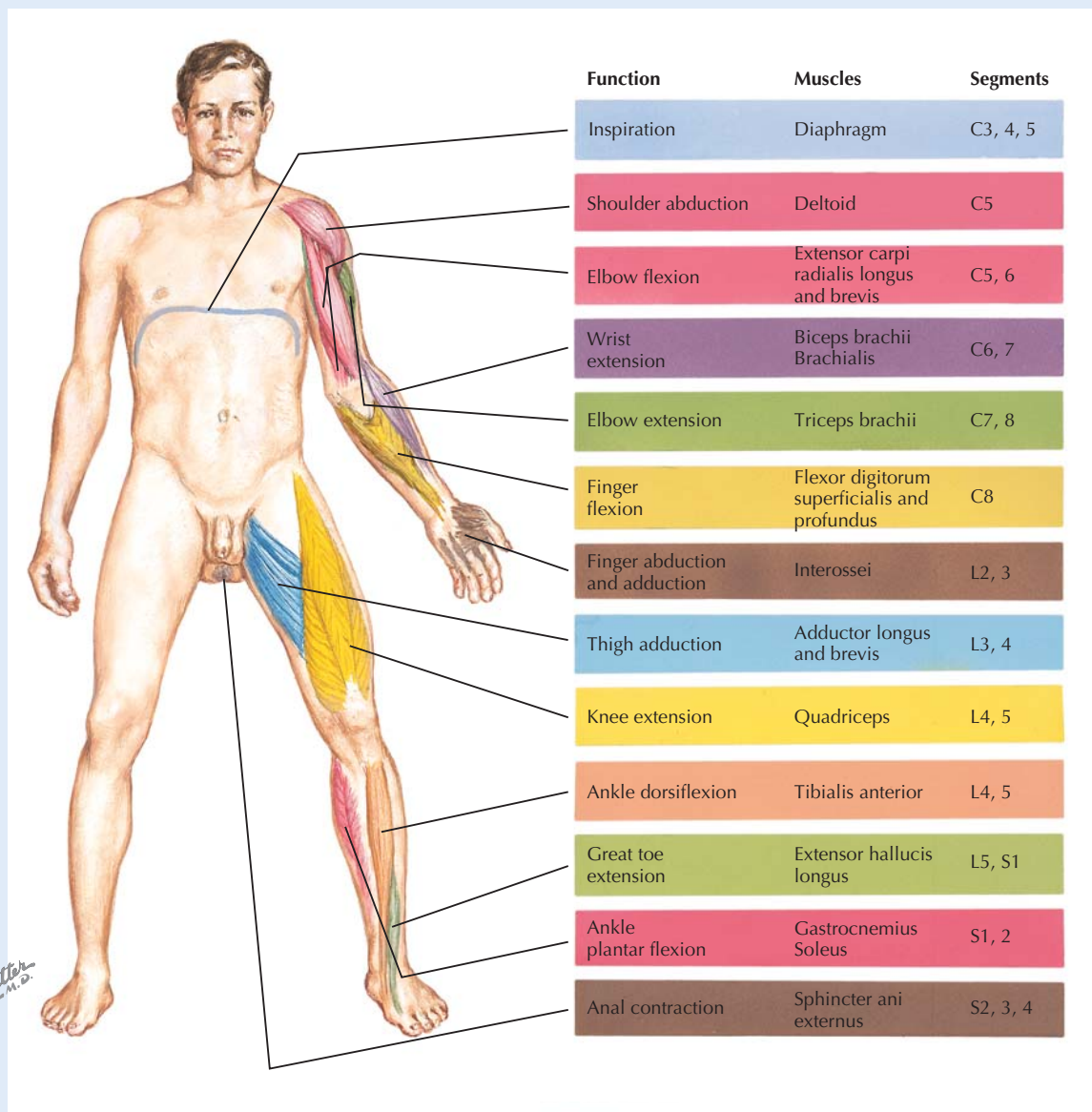
Figure 4.9 Somatic Component of the Peripheral Nervous System The peripheral nervous system can be subdivided into somatic and autonomic components. The somatic nervous system contains motor nerves and sensory nerves innervating skin and muscle. The soma (cell bodies) of motor nerves and sensory nerves are located in the gray matter of the anterior horn of the spinal cord and in the dorsal root ganglia, respectively.

CLINICAL CORRELATE

Spinal Cord Injury

Injury to the spinal cord results in deficits in motor and sensory function below the point of injury. The most common cause of spinal cord injury is trauma; other causes include neurodegenerative diseases as well as tumors and vascular diseases affecting the spine. Spinal cord injury is classified as complete, when no neurological function is retained beyond the point of damage, or incomplete, when some sensory or motor function remains. Effects of spinal cord damage correspond to the level of injury, with increasing severity at higher levels:

- Sacral and lumbar injuries are associated with problems in sexual, bladder, and bowel function, as well as motor and sensory deficits in the lower limbs.
- Thoracic injuries are associated with paraplegia (paralysis of the lower body).
- Cervical injuries result in some degree of quadriplegia (paralysis below the neck), depending on the level of injury. Patients with injuries at spinal cord level C3 or above require mechanical ventilation.



Motor Impairment Related to Level of Spinal Injury

CHAPTER 5

Sensory Physiology

Sensory systems allow us to maintain homeostasis and respond to the external environment in an adaptive manner. Much of the awake brain is devoted to transducing, encoding, sensing, and eventually perceiving and reacting to or trying to make sense of both our external and internal environments. An important task of the brain is to filter out the large amounts of incoming information to prevent information overload. During sleep, the incoming sensory information is drastically reduced as the brain continues to try to make sense of the stored information and “reset” itself for the next day’s influx of sensory inputs.

SENSORY RECEPTORS

Specialized receptors detect stimuli of various types, including visual, olfactory, gustatory (taste), auditory, and somatosensory (mechanical, thermal, and painful) stimuli. Stimulation of sensory receptors results in the opening or closing of ion channels and generation of a **receptor or generator potential** (a membrane potential change in the receptor). When threshold is reached, the stimulus has been transduced into an electrical signal, and information is transmitted via afferent pathways to the central nervous system (CNS), which integrates such information and transmits signals back to effector systems. The **somatosensory system** includes **mechanoreceptors**, **thermal receptors**, and **nociceptors** (pain receptors) that respond to stimuli in the skin or at the body surface. The term **somatovisceral sensory system** is also used, because such receptors are also located in visceral organs, muscle, and joints.

Several types of sensory receptors are found in the skin (and other tissues); these can be differentiated by the stimuli to which they respond, as well as by the degree of adaptation they exhibit to intensity, duration, and change in stimulus (Fig. 5.1):

- **Meissner’s corpuscles** are mechanoreceptors located in the dermal papillae, especially in fingertips, palms, soles, lips, face, tongue, and genital skin (nonhairy skin). These are rapidly adapting receptors with small receptive fields that allow point discrimination and detection of low-frequency stimuli such as flutter.

- **Pacinian corpuscles** are rapidly adapting mechanoreceptors that detect pressure and vibration. The lamellated capsule of these receptors allows them to respond specifically to rapid changes in pressure as well as vibration, as opposed to continual pressure or slow changes in pressure (Fig. 5.2).
- **Merkel’s disks** are slowly adapting mechanoreceptors that respond through small receptive fields to pressure and touch, particularly to indentation of the skin.
- **Hair follicle receptors** consist of nerve endings wrapped around the base of hair follicles. These rapidly adapting mechanoreceptors detect movement across the surface of the skin.
- **Ruffini’s corpuscles** are mechanoreceptors found in the dermis and in joints; they respond to stretch and are slowly adapting.
- **Thermal receptors** are sensory nerves with endings in the skin that respond to temperature. They are subdivided into warm and cold receptors. Both are slowly adapting.
- **Nociceptors** are sensory nerves with endings in skin, cornea, muscle, joints, and visceral endings. Various types of nociceptors respond to temperature and mechanical and chemical stimuli.
- **Muscle and joint receptors** include muscle spindles and Golgi tendon organs. These receptors are important in proprioception and coordination of motor activity and are considered in more detail later (see “The Motor System”).



The receptive field of a sensory neuron is the area over which a stimulus will trigger firing of the neuron. A rapidly adapting receptor is one that responds quickly to a stimulus but adapts rapidly to a constant stimulus, returning to its normal firing rate.

Transduction of Somatosensory Signals

Somatosensory (pain, touch, pressure, temperature) signals originating below the head and in the head are ultimately conveyed to the primary somatosensory area, located in the post-central gyrus of the parietal lobe of the cerebral cortex. The

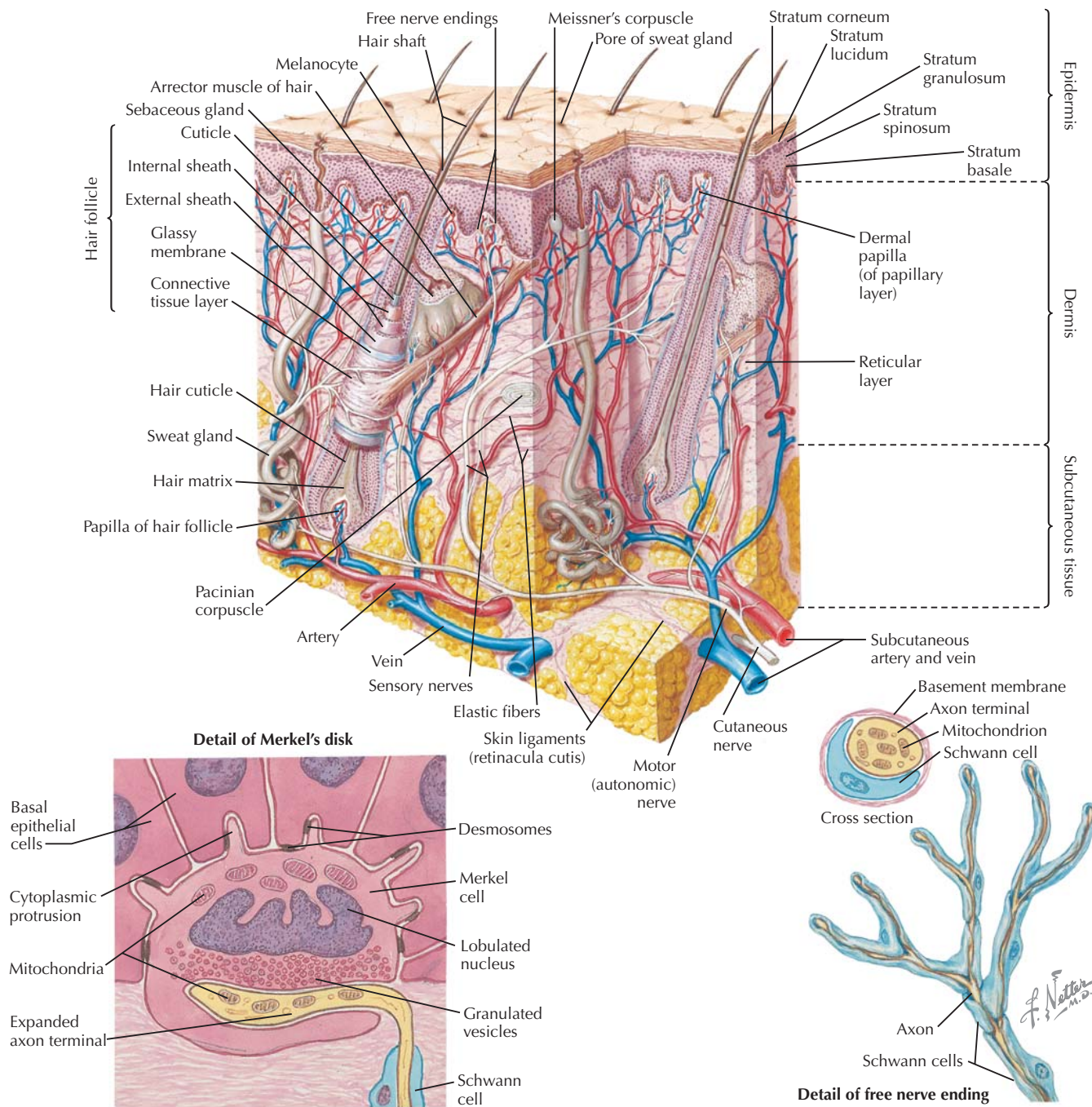


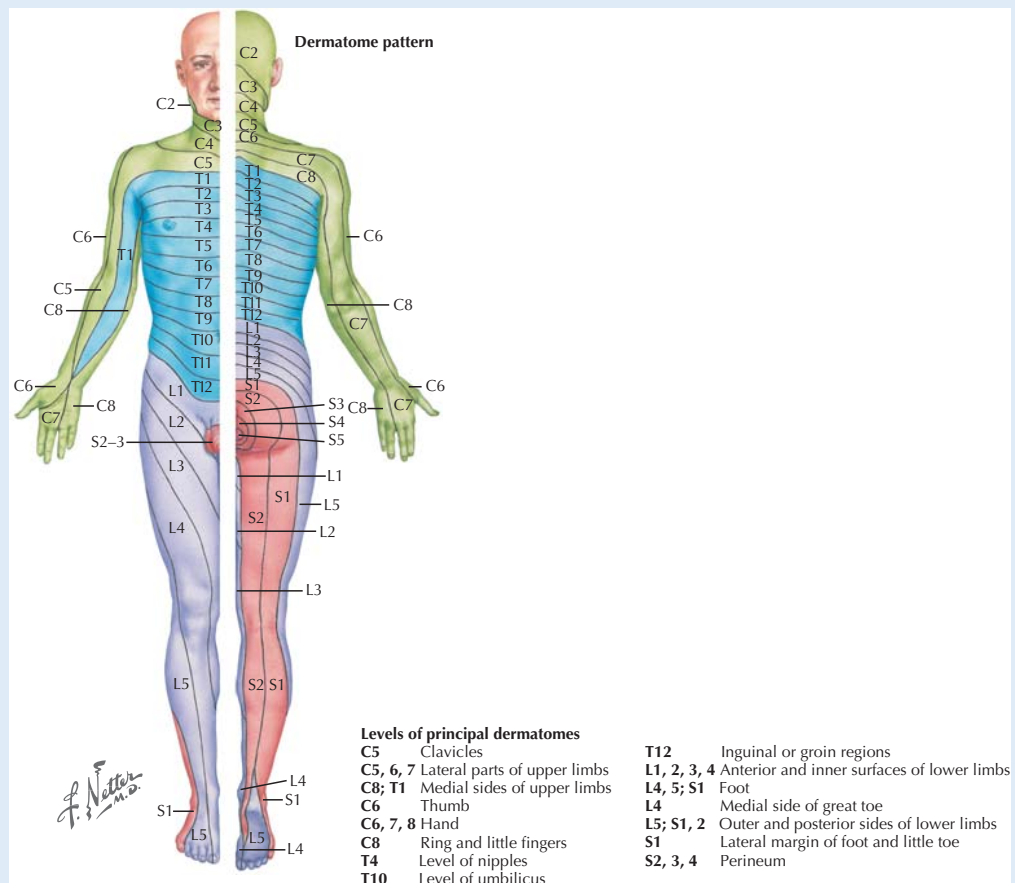
Figure 5.1 Skin and Cutaneous Receptors Mechanoreceptors, thermoreceptors, and nociceptors transduce the effects of touch, temperature, and painful stimuli into neural signals. Within the skin, mechanoreceptors known as Meissner's corpuscles, pacinian corpuscles, and Merkel's disks respond to various types of mechanical stimulation. Temperature and painful stimuli are detected by free nerve endings.

CLINICAL CORRELATE

Dermatomes

Dermatomes are areas of the skin associated with a specific pair of dorsal nerve roots. In other words, sensory signals originating in the skin within a dermatome are largely transmitted to the spinal cord through one of a specific pair of afferent sensory nerve roots, although there is significant overlap between adjacent dermatomes. Clinically, dermatomes are useful in identifying the site

of spinal nerve damage, based on the region of skin in which sensation is affected. For example, sensations in the feet are conveyed to the spinal cord through nerve roots at L4, L5, or S1, depending on the area of the foot, with some overlap in regions. In shingles (herpes zoster), the herpes virus usually resides dormant in nerve cell bodies of the dorsal root ganglia, but when active, will specifically affect the skin in the corresponding dermatome on one or both sides (depending which ganglia are infected).



afferent somatosensory signals originating below the head are conveyed to the dorsal root ganglia and subsequently through the spinothalamic and spinoreticular tracts of the anterolateral system, ultimately reaching the primary somatosensory cortex (Fig. 5.3). Signals involved in proprioception, as well as signals generated by vibratory and tactile stimuli, are carried through the fasciculus gracilis and fasciculus cuneatus to the ventral posterolateral nucleus (VPN) of the thalamus. The lateral cervical system carries some proprioceptive, vibratory, and tactile signals as well. These pathways reach synapses in the thalamus and before extending to the cerebral cortex.

In the head, touch, pressure, pain, and temperature result in afferent nerve traffic to nerve cell bodies in the trigeminal

(semilunar) ganglia of the trigeminal nerve. Signals involved in proprioception are carried to neuronal cell bodies in the mesencephalic nucleus of the trigeminal nerve (CN V). Projections in this system are mainly to the contralateral ventral posterolateral nucleus (VPN) of the thalamus, subsequently reaching the primary somatosensory cortex (postcentral gyrus) (Fig. 5.4).

THE VISUAL SYSTEM

The visual system performs the remarkable task of detecting light stimuli between 400 and 700 nanometers in wavelength, transducing the stimuli into electrical signals, and transmitting

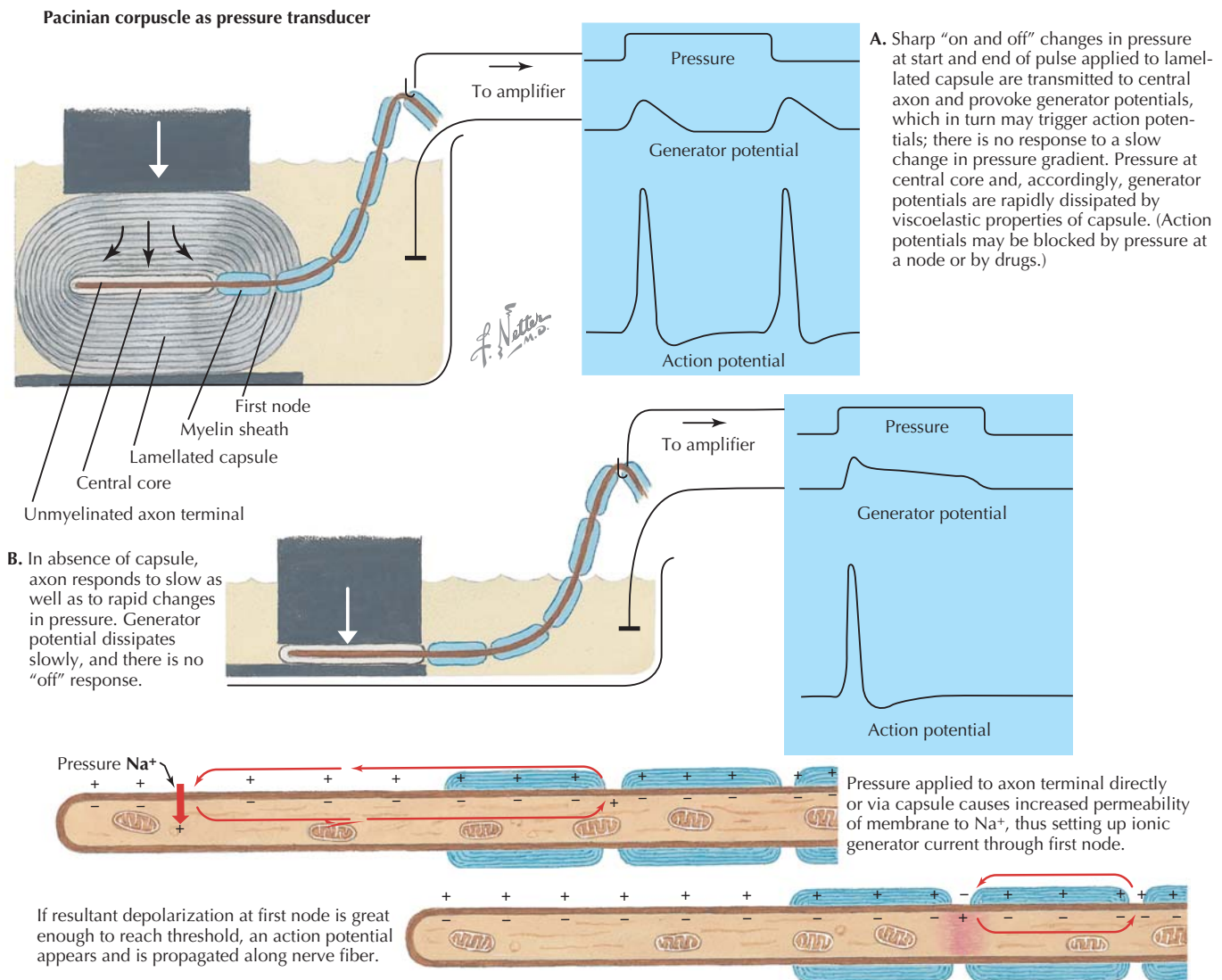


Figure 5.2 Pacinian Corpuscle Pressure and vibration are transduced into electrical potential changes by pacinian corpuscles, which are rapidly adapting mechanoreceptors. The lamellated capsule surrounding the axon terminal dissipates slow pressure changes, and thus, the axon responds only to rapid changes in pressure (A), in contrast to an axon lacking a capsule (B). The action potentials produced are carried through afferent fibers to the dorsal root ganglia (see Fig. 4.9).

these signals to the CNS, where they are used to construct a three-dimensional representation of objects in the visual field that includes color and brightness and detects motion.

Light passes through the cornea of the eye, through the pupil (the circular opening of the iris), and is then focused by the lens onto the retina (Fig. 5.5). Within the retina, photoreceptors known as rods and cones contain the pigment **rhodopsin**, which absorbs photons. Color perception is mainly mediated by cones, whereas rods are sensitive photoreceptors adapted to perceive light at low intensity. The visual fields of the two eyes overlap, with the center being projected, upside-down, on the sensitive macular zone of the retina (Fig. 5.6). Within

the macula, the **fovea** is the region of greatest visual acuity; it contains only cones.

Reception of light by the photoreceptors initiates a sequence of events that results in reduced Na^+ permeability and hyperpolarization of the photoreceptor cell membrane. This results in inhibition of the release of either inhibitory or excitatory neurotransmitters at synapses between photoreceptors and **bipolar** and **horizontal cells**. Ultimately, the generated signals are transmitted by the **ganglion cells**, whose axons form the **optic nerves**. Nerve fibers originating in the nasal hemiretina (nasal half of the retina) of each eye cross at the **optic chiasm** (the site of partial crossing of the two optic nerves), while

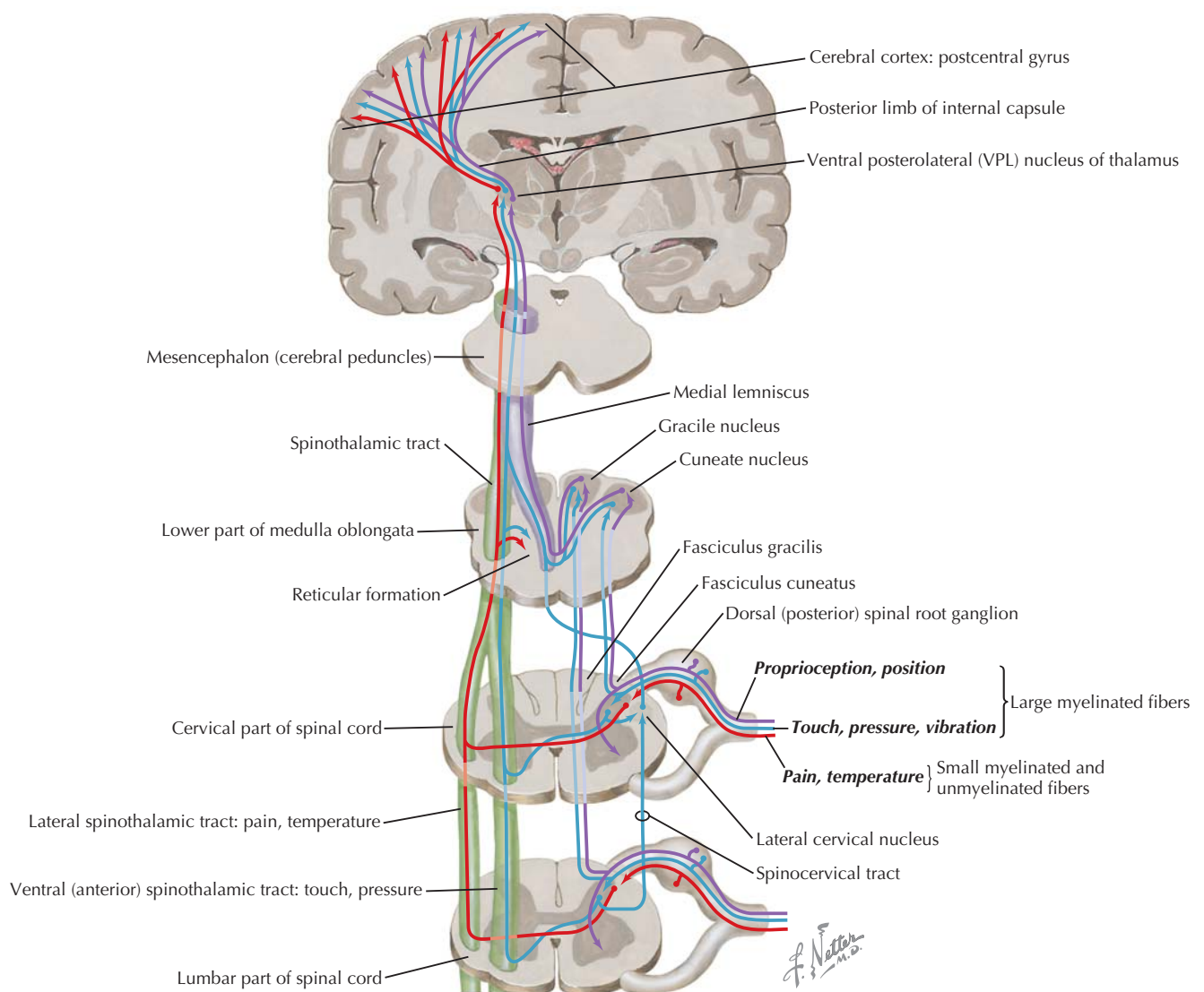


Figure 5.3 Somesthetic System of the Body The primary somatosensory cortex (postcentral gyrus) receives information regarding proprioception, touch, pressure, vibration, pain, and temperature originating at receptors below the head through the illustrated tracts by way of the thalamus. Pain-, temperature-, and pressure-related signals are conducted through the spinothalamic and spinoreticular tracts of the anterolateral system (red and blue lines). The fasciculus gracilis and fasciculus cuneatus, as well as the lateral cervical system, convey proprioception-, touch-, and vibration-related signals (blue and purple lines).

fibers from the temporal hemiretina remain in the ipsilateral tract (same side). The optic tracts carry the signals to synapses at the **lateral geniculate nuclei**, from which signals are transmitted to the visual cortex in the occipital lobe of the cerebral cortex.

THE AUDITORY SYSTEM

The **auditory system** detects sound, transduces it into electrical signals, and transmits it to the brain, where the complex

task of interpretation of sound, for example as speech, takes place. The process begins at the external ear, which directs sound to the tympanic membrane (eardrum) (Fig. 5.7). This membrane separates the external ear from the air-filled middle ear, where sound is amplified. Vibration of the tympanic membrane is conducted to the malleus, incus, and stapes, the small bones of the middle ear known collectively as the **ossicles**. The stapes is connected to the **oval window**, one of two membranous “windows” between the middle ear and inner ear. Thus, vibration of the ossicles is transmitted to the inner ear by the stapes.

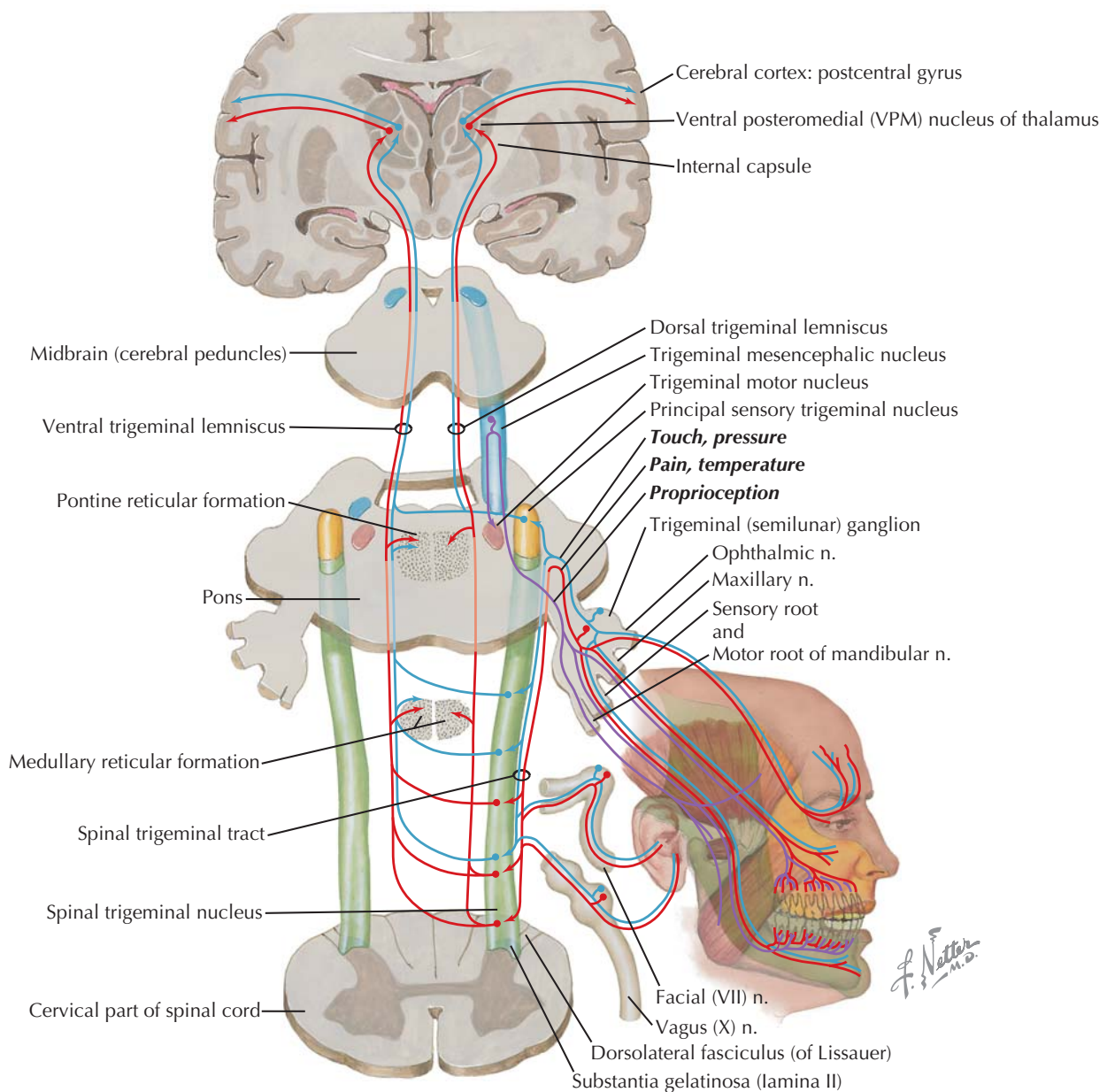


Figure 5.4 Somesthetic System of the Head The primary somatosensory cortex (postcentral gyrus) receives information regarding proprioception, touch, pressure, pain, and temperature originating at receptors in the head through the illustrated tracts by way of the thalamus. Proprioceptive signals are carried by nerve fibers (*purple lines*) originating in cell bodies in the mesencephalic nucleus of the trigeminal nerve (CN V), whereas touch, pressure, pain, and temperature produce signals carried by nerve fibers (*blue and red lines*) to cell bodies in the trigeminal ganglion of CN V.

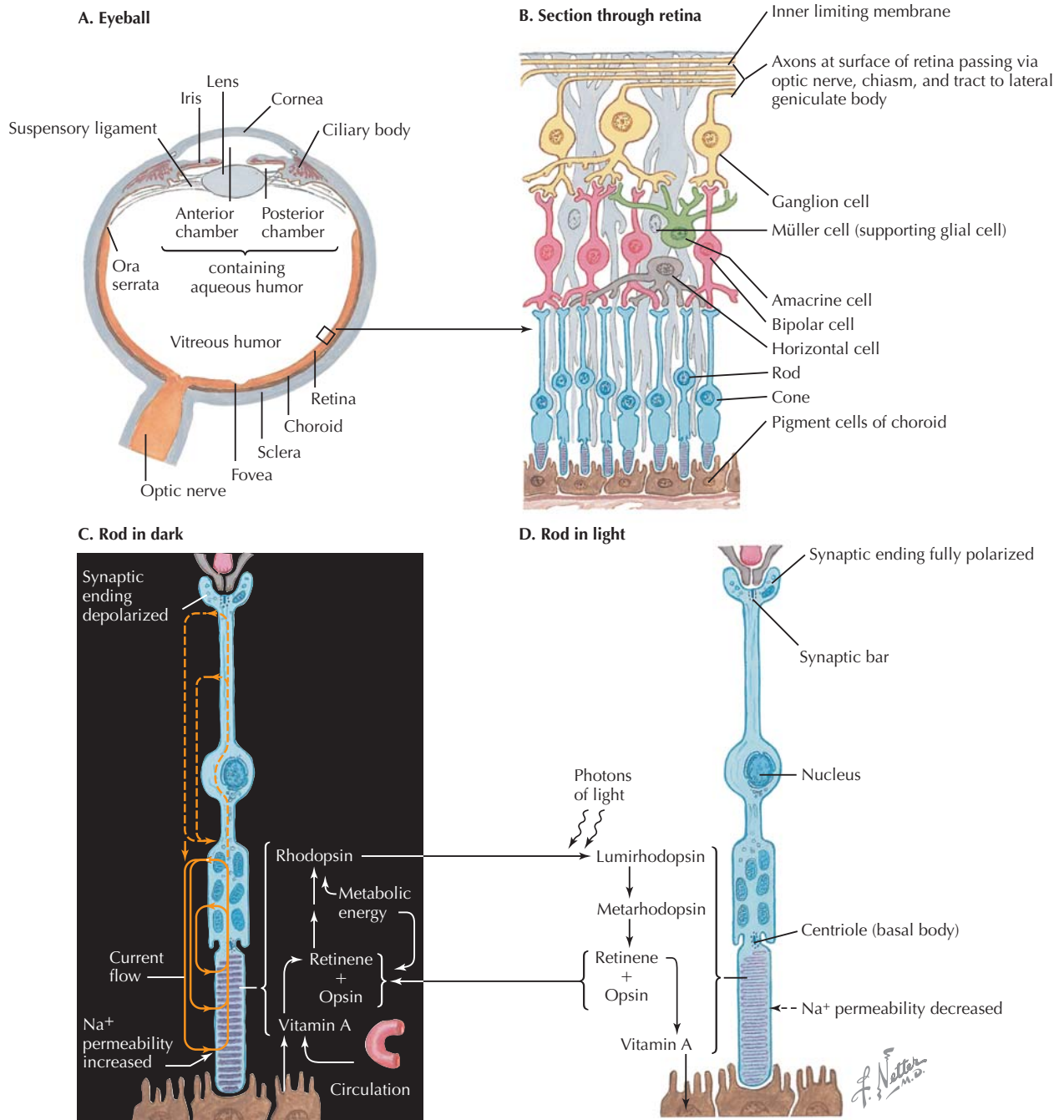


Figure 5.5 Visual Receptors Light enters the eye through the cornea and lens (A). Light focused onto the retina by the lens is absorbed by rhodopsin in photoreceptors (rods and cones, B). In light, the cGMP second messenger system is activated, resulting in closure of membrane Na^+ channels, hyperpolarizing the cell (illustrated for rod cells in D). Subsequent inhibition of neurotransmitter release at synapses with bipolar and horizontal cells is the next step in the transmission of light-generated signals to the optic nerve. In dark, the rod or cone is depolarized (C).

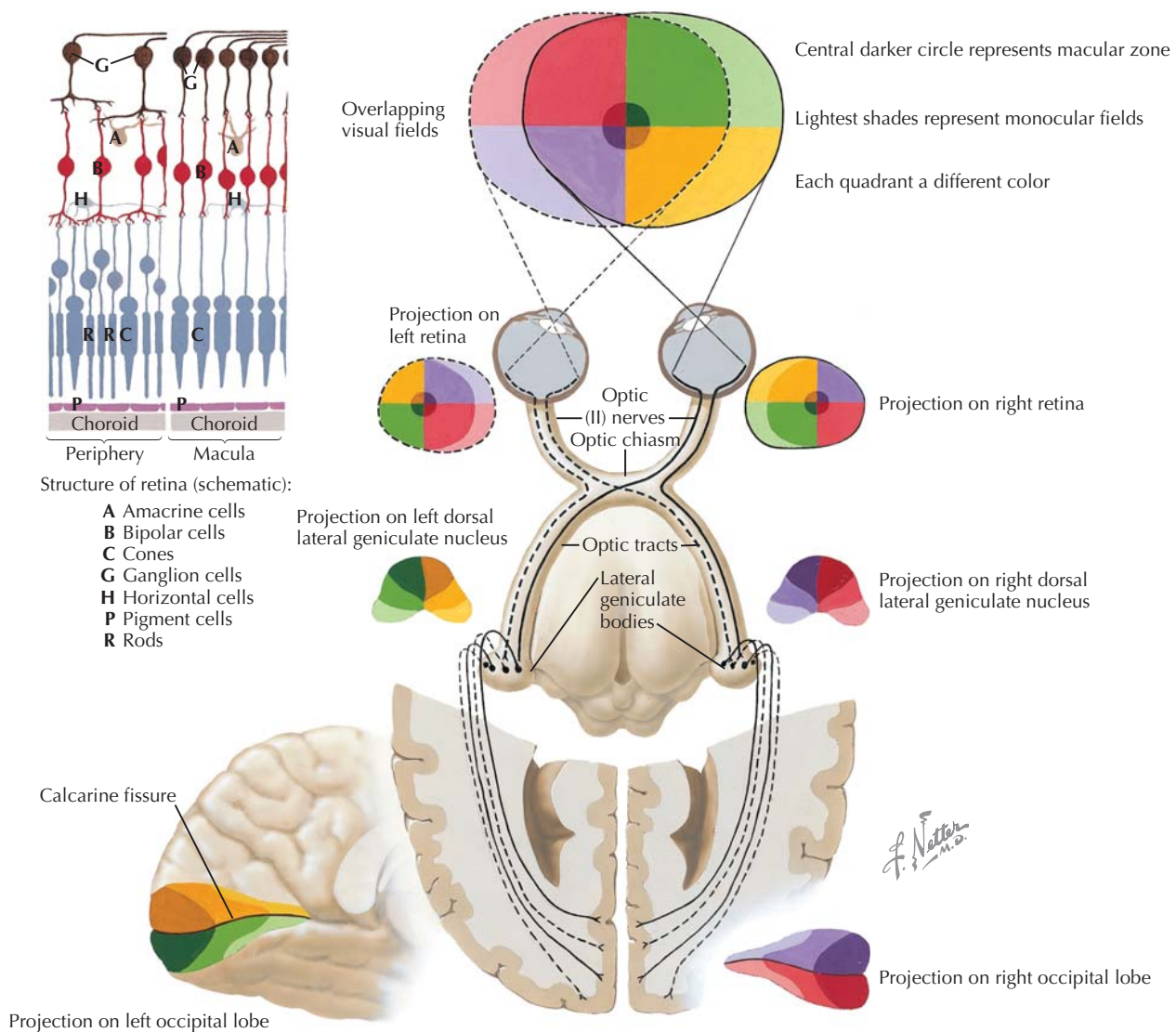


Figure 5.6 Retinogeniculostriate Visual Pathway Rods and cones detect light in the retina, with cones specifically mediating color perception and rods detecting light. In the central macular zone of the retina, only cones are found; this is the region with greatest acuity (*top left*). Axons of ganglion cells carry afferent signals through the optic nerve. At the optic chiasm, fibers originating in the nasal half of the retina cross to the contralateral tract, whereas fibers from the temporal half remain in the ipsilateral tract. Fibers project to synapses in the lateral geniculate nucleus (visual fields are inverted at this point). Signals are conveyed from the lateral geniculate nucleus to the primary visual cortex.

The fluid-filled inner ear is a complex structure that includes the vestibule, the cochlea, and the semicircular canals. The vestibule and cochlea consist of a **membranous labyrinth** within a **bony labyrinth** (Fig. 5.8). The membranous labyrinth of the cochlea includes three ducts known as the scala vestibuli, scala tympani, and scala media (cochlear duct). The fluid of the scala media, the **endolymph**, resembles intracellular

fluid in its composition, whereas the **perilymph** of the other two ducts is similar to other extracellular fluids.

Transduction of Sound into Electrical Signals

The **spiral organ of Corti** is the site of transduction of sound into electrical signals. Mechanoreceptors known as **hair cells**

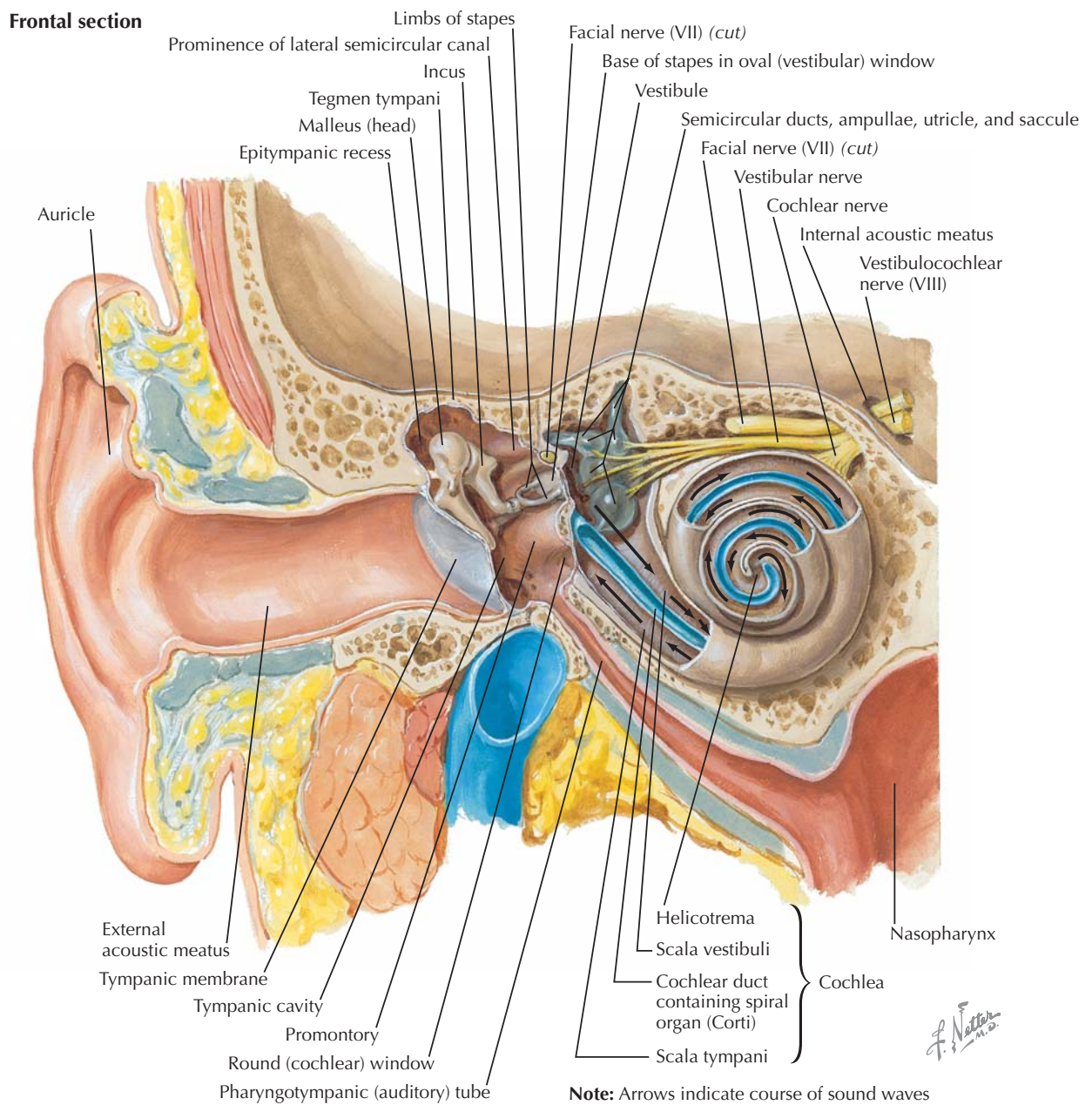
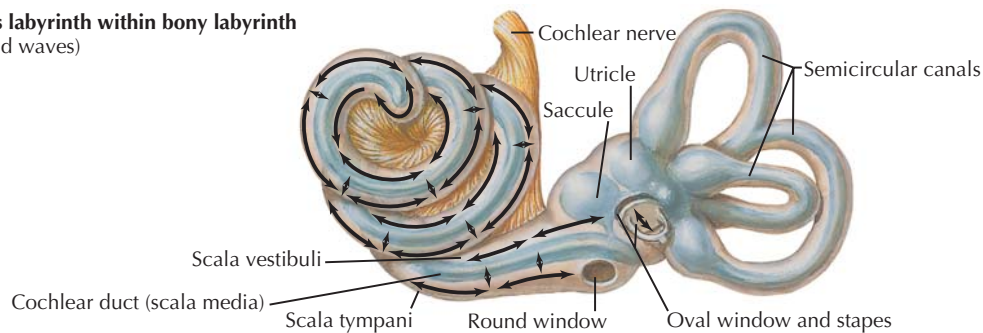


Figure 5.7 Peripheral Pathways for Sound Reception The process of auditory perception begins when sound reaches the auricle (external ear) and is directed through the external acoustic meatus to the tympanic membrane, causing it to vibrate. The vibration is transmitted by the bones of the air-filled middle ear (malleus, incus, and stapes) to the fluid-filled inner ear via the oval window. Within the inner ear, the organ of Corti of the cochlea contains auditory receptors (see Fig. 5.8).

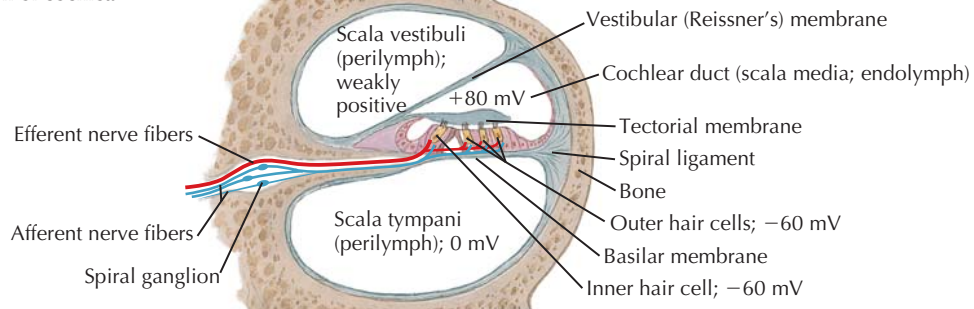
have **stereocilia** with their distal ends extending to and embedded in the **tectorial membrane** (see Fig. 5.8). When sound is conducted into the fluid of the inner ear, vibration of the basilar membrane results in bending of these stereocilia. The endolymph is unusual in terms of its electrical potential (approximately +80 mV relative to the perilymph), resulting

in a charge difference across the apical membrane of approximately 170 mV. Bending of the stereocilia results in changes in cation conductance of the apical cell membrane of hair cells; bending in one direction produces hyperpolarization, and bending in the other direction causes depolarization. As a result, there is an oscillatory release of neurotransmitter at

A. Membranous labyrinth within bony labyrinth
(path of sound waves)



B. Section through turn of cochlea



C. Spiral organ of Corti

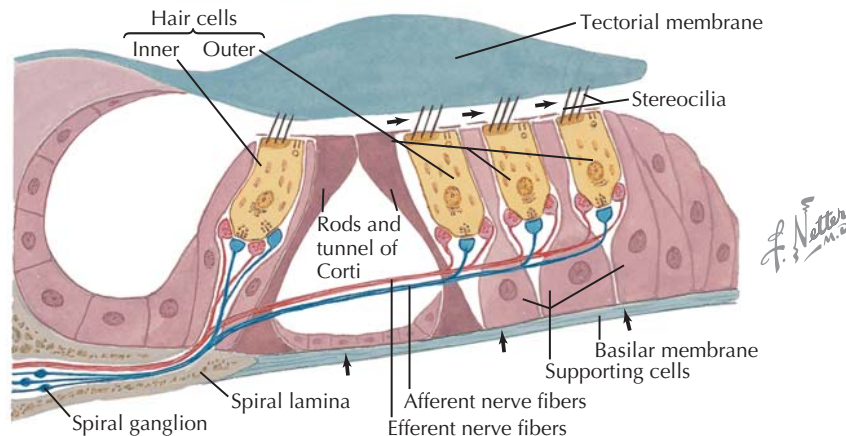


Figure 5.8 Cochlear Receptors Sound waves reaching the inner ear travel through the membranous labyrinth of the cochlea (A), which consists of three ducts (B). When vibration caused by sound reaches the cochlea of the inner ear, periodic displacement of the basilar membrane relative to the tectorial membrane causes bending of stereocilia and depolarizes hair cells within the spiral organ of Corti, resulting in transduction of sound into neural signals (C).

the synapses between hair cells and afferent nerve fibers, with depolarization of hair cells producing increased neurotransmitter release and hyperpolarization causing reduced release. When threshold is reached in the afferent nerve fibers (during depolarizations of the hair cells), action potentials are produced. Within regions of the organ of Corti, the amplitude by which the basilar membrane is repeatedly displaced as it vibrates is dependent on sound frequency. Thus, the organ of Corti is tonotopically organized, with high-frequency sounds causing the largest displacement in the basal region of the cochlea, and low-frequency sounds causing the largest dis-

placement in the apical region (Fig. 5.9). As a result, cochlear nerve fibers respond to different frequencies of sound, according to the **characteristic frequency** of the fiber (the sound frequency of lowest intensity that produces a response in the fiber).

Impulses pass through the **auditory nerve** to the brainstem, where neurons compute various sound parameters (see Fig. 5.9). For example, the **superior olivary complex** detects differences in timing and intensity of sound signals originating from the two ears, allowing us to detect the location of a

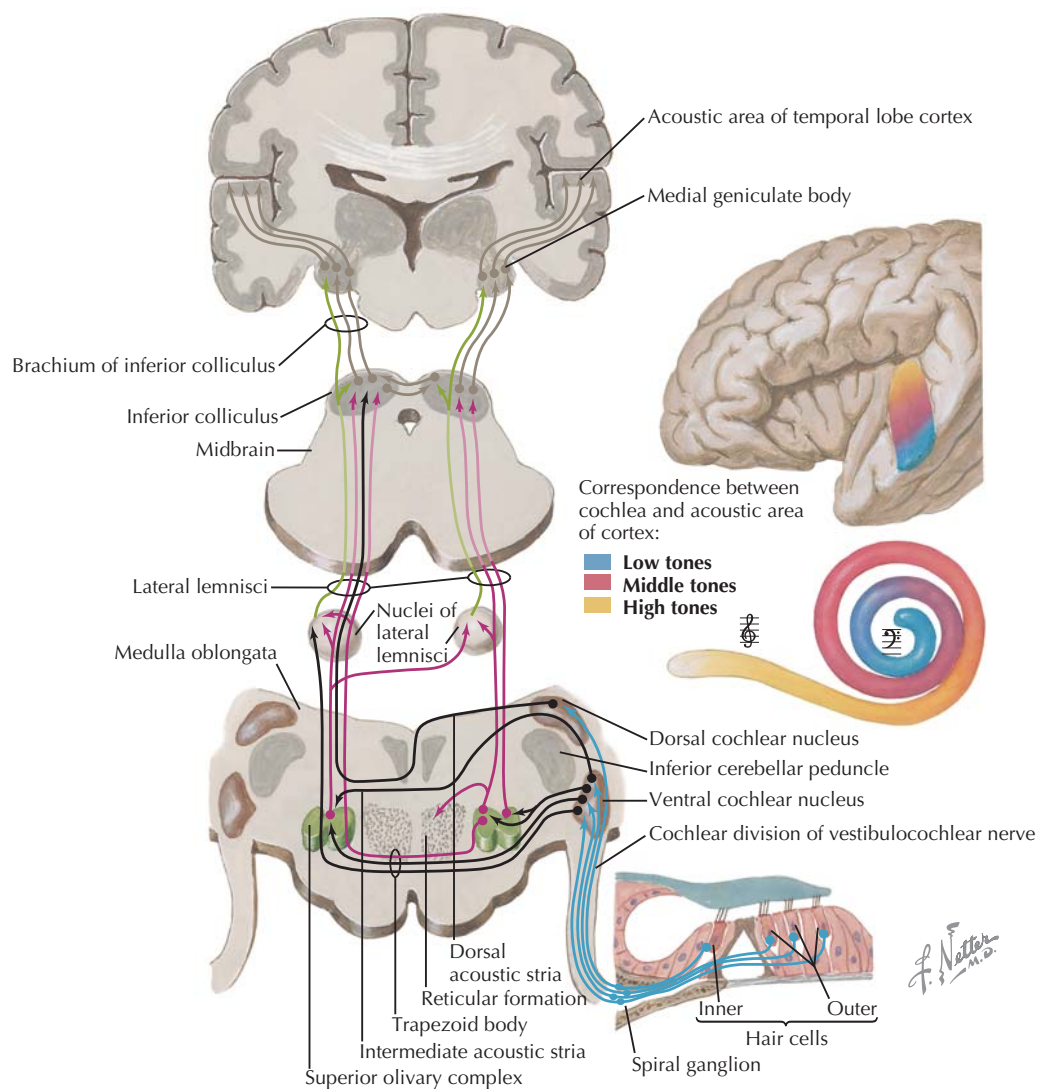


Figure 5.9 Auditory Pathways Electrical signals generated by hair cells in the organ of Corti are conveyed to the dorsal and ventral cochlear nuclei of the medulla, which project to the lateral lemniscus. After further relays, the pathway leads to the medial geniculate bodies of the thalamus, which sends projections to the primary auditory cortex.

sound source. Once the information reaches the thalamus (**medial geniculate body**), it is further processed and relayed to the amygdala and **auditory areas** of the temporal lobe. Within these areas, there is correspondence of tonotopic organization to that within the cochlea. Other parameters, such as amplitude and pitch of a sound, are also mapped there.

THE VESTIBULAR SYSTEM

In addition to its role in audition, the inner ear has an important role in balance and equilibrium through its **vestibular**

apparatus, which consist of the **semicircular canals** and the **otolithic organs** (the **utricle** and the **sacculle**) (Fig. 5.10). These structures detect angular and linear acceleration of the head, and thus play an important role in proprioception. Specifically, the three semicircular canals are oriented perpendicularly to each other and are therefore able to detect angular acceleration in any plane; the otolithic organs detect linear acceleration.

The sensory tissues of the semicircular canals are found in the **ampullary crest** (**crista ampullaris**). Hair cells in this structure, like those of the organ of Corti, have apical stereocilia, but in this case, each hair cell also has a single large

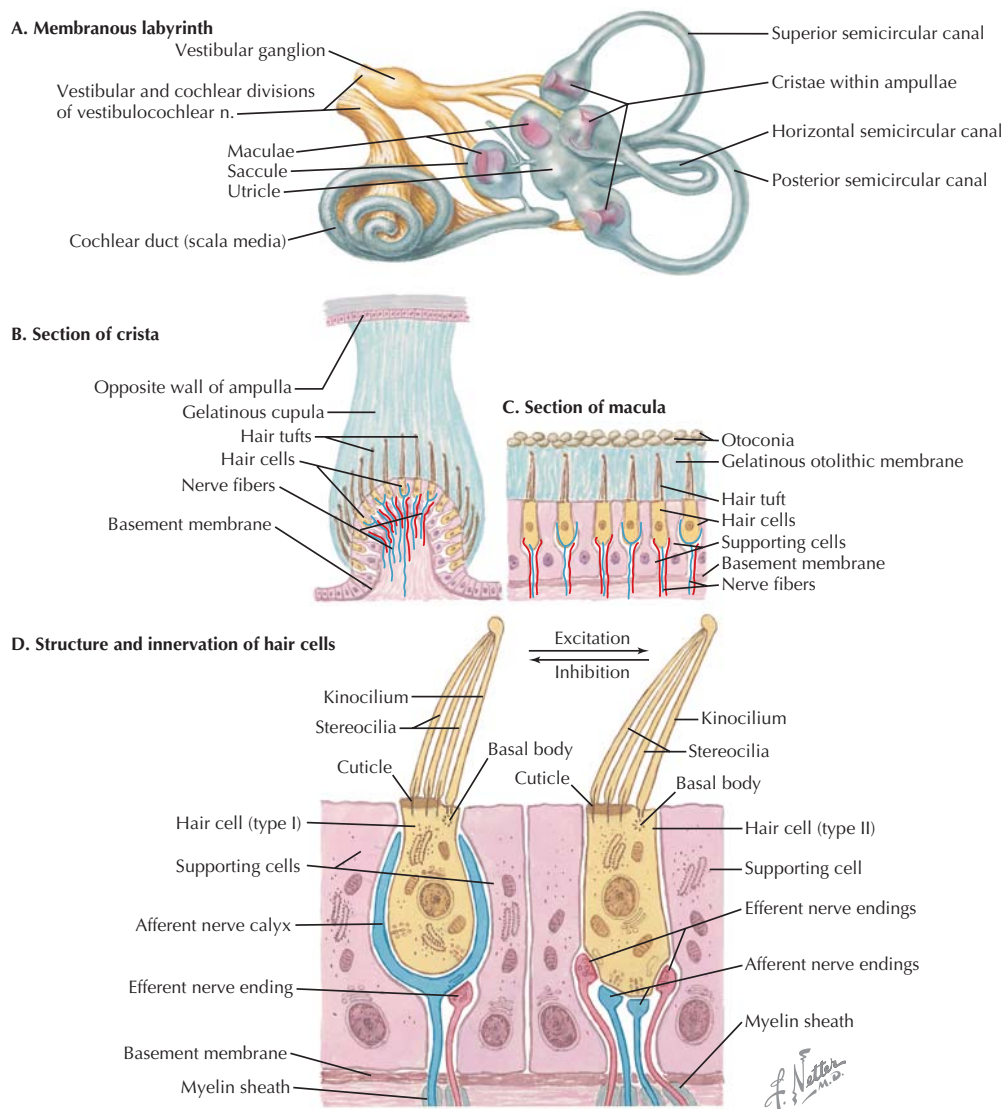


Figure 5.10 Vestibular Receptors The semicircular canals and the otolithic organs (the utricle and saccule) constitute the vestibular apparatus of the inner ear, which has a critical role in balance and equilibrium (A). The otolithic organs detect linear acceleration of the head, whereas the three perpendicular semicircular canals respond to angular acceleration. Sensory hair cells in the crista of the semicircular canals (B) and macula of the otolithic organs (C) respond to fluid (endolymph) movement during linear and angular acceleration (D).

cilium known as the **kinocilium**. A gelatinous mass, the **cupula**, spans and occludes the ampulla at the ampullary crest. The semicircular canals are filled with endolymph, and during angular acceleration of the head, pressure against the cupula causes bending of the cilia. Due to the perpendicular orientation of the three semicircular canals, the pattern of pressure changes in the canals is dependent on the direction of movement of the head. When cilia bend toward the kinocilium, the membrane potential becomes depolarized (due to increased conductance to cations); bending in the opposite direction causes hyperpolarization. Depolarization results in

release of neurotransmitter, whereas hyperpolarization reduces neurotransmitter release by the hair cells. Changes in neurotransmitter release alter the firing rate of afferent nerve fibers ending on the hair cells. Impulses are carried by axons of these primary afferent nerves through cranial nerve VIII to the vestibular nuclei of the pons (Fig. 5.11). Subsequent processing involves both ascending and descending pathways that carry the information through secondary axons to the spinal cord, cerebellum, reticular formation, extraocular muscles, and cortex (by way of the thalamus).

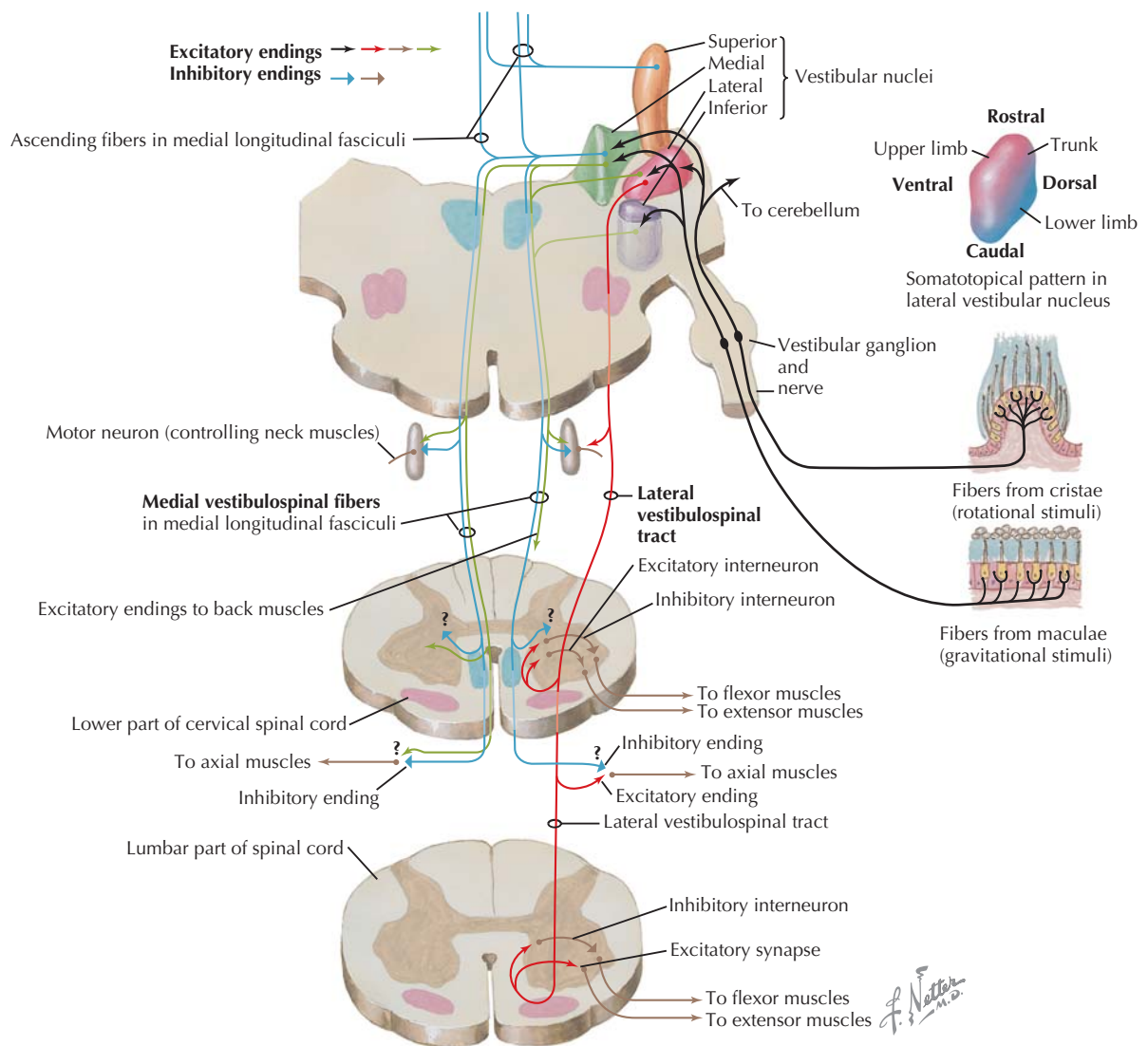


Figure 5.11 Vestibulospinal Tracts Sensory signals from the vestibular apparatus are conveyed to the pontine vestibular nucleus, and subsequently, through secondary axons, to the spinal cord, cerebellar vermis, reticular formation of the brainstem, extraocular muscles, and the cerebral cortex (by way of the thalamus). Vestibular sensory input is utilized to maintain balance and posture and in positioning of the head.

In the standing position, the utricle senses horizontal acceleration, whereas the saccule is sensitive to vertical acceleration. This is a result of the orientation of the sensory tissues of the otolithic organs, the **maculae** (see Fig. 5.10). The macula is oriented horizontally in the utricle and vertically in the saccule and contains hair cells like those in the crista of the semicircular canals. A gelatinous **otolithic membrane** lies above the macula. Linear acceleration produces pressure changes in endolymph-filled otolithic organs, shifting the macula and bending hair cells. Again, depolarization or hyperpolarization results, affecting transmitter release, afferent nerve firing, and



Umami (Japanese for “savory”) has long been recognized as a basic taste classification in Japan. It is used to describe the savory taste of protein-rich dishes such as those containing meat. The recent discovery of a specific taste receptor that binds glutamates, which are prevalent in protein-rich food, has led to the recognition of the “umami taste receptor,” along with the traditional sweet, bitter, salty, and sour receptors on taste buds. The flavor-enhancing quality of monosodium glutamate (MSG) can be ascribed to stimulation of the umami receptor.

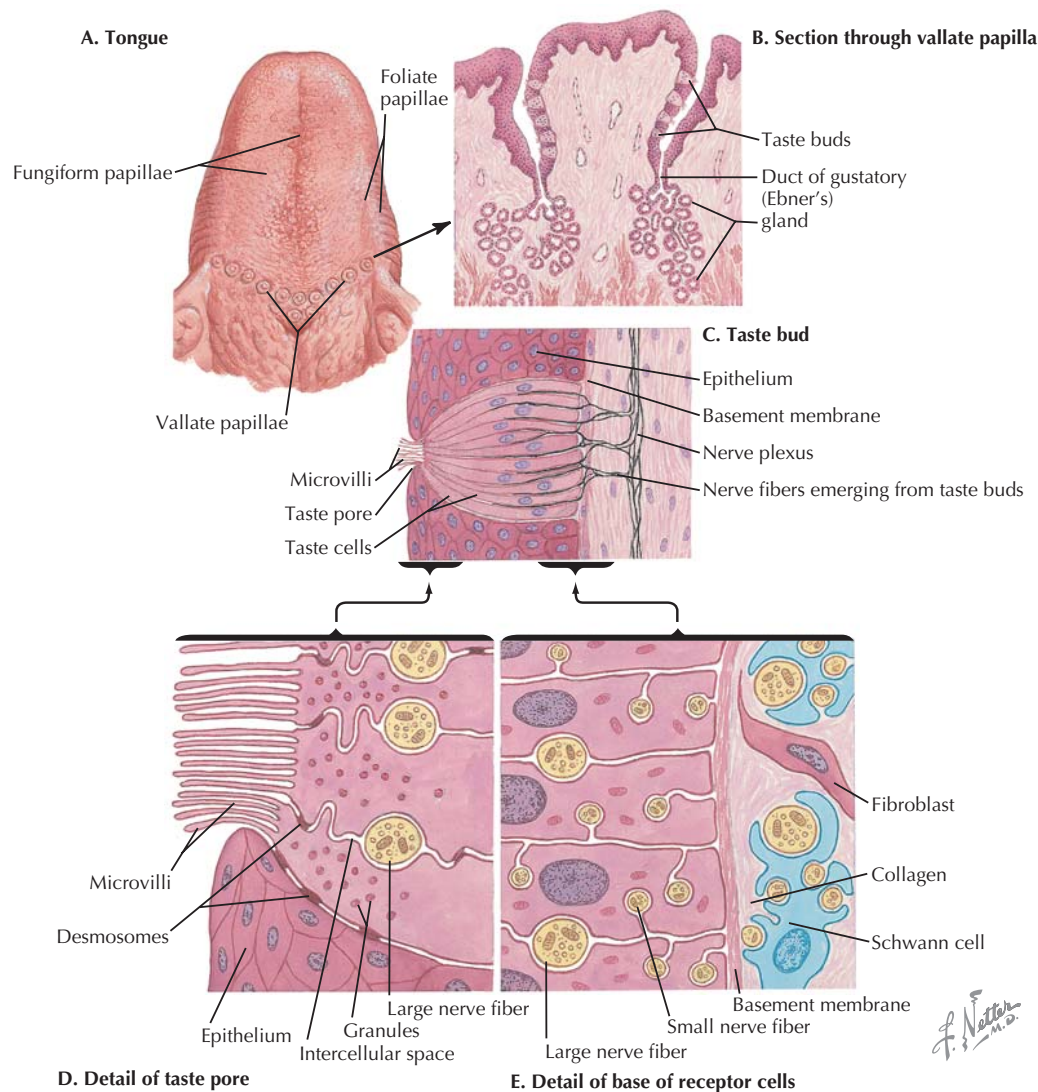


Figure 5.12 Taste Receptors Taste receptors are found mainly in the taste buds of the three types of papillae of the tongue (A and B), but also in taste buds on the palate, larynx, and pharynx. Goblet-shaped taste buds (C) comprise taste cells with microvilli protruding through the taste pore (D). Nerve fibers are found in close association with hair cells (D and E). Taste cells, through receptors, respond to chemical signals (taste), ultimately producing depolarization and action potentials in afferent nerve fibers. At least five types of taste receptors exist: sweet, salty, sour, bitter, and umami (savory).

transmission of signals to the brainstem (see Fig. 5.11). By virtue of the orientation of the maculae of the two otolithic organs, linear acceleration in any direction can be detected.

CHEMICAL SENSES

Taste Cells

Gustation (taste) and olfaction (smell) are two chemical senses involved in detecting chemical stimuli in the environ-

ment and the substances we ingest. Traditionally, tastes have been classified as sweet, bitter, salty, and sour. More recently, a fifth classification, **umami** (savory) has become widely accepted (see preceding box). **Taste cells** (also called taste receptor cells or gustatory receptor cells) are found mainly in the **taste buds** of the papillae of the tongue, but also in taste buds of the palate, larynx, and pharynx. Three types of papillae are recognized, based on anatomical structure: the fungiform, foliate, and circumvallate papillae (Fig. 5.12). Taste buds within the papillae consist of groups of taste cells, along with supporting cells and basal cells; the latter differentiate to continuously replace taste cells. Individual taste cells respond

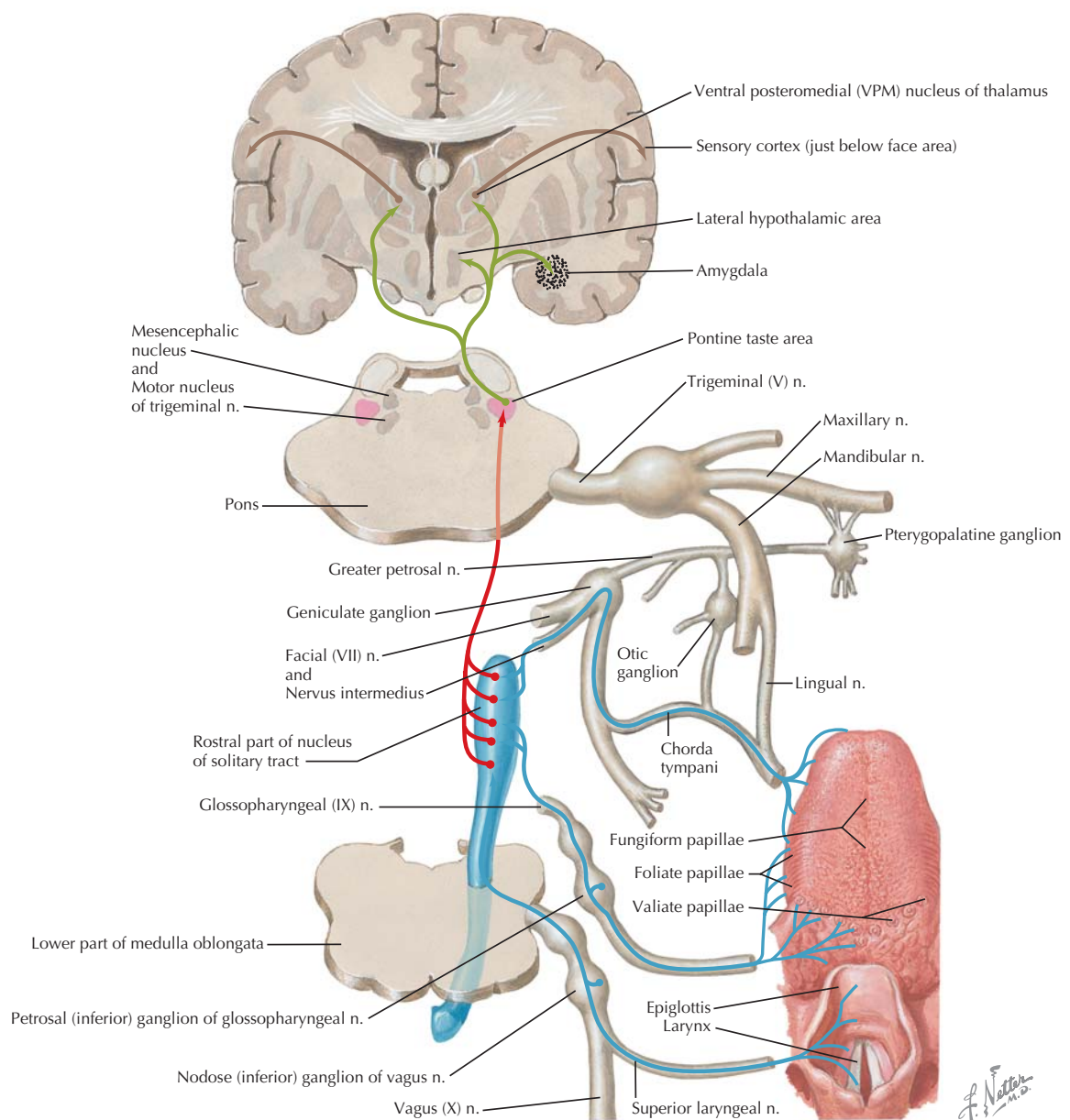


Figure 5.13 Taste Pathways Regions of the tongue are innervated by cranial nerves VII, IX, and X. These pathways transmit afferent signals to the medulla (specifically, the nucleus tractus solitarius), from which they are conveyed to other brain regions, eventually reaching the sensory cortex.

through receptors to all of the fundamental taste types, although they may respond best to one type. Microvilli of taste cells recognize chemical signals in saliva, resulting in activation of G protein–coupled signal transduction mechanisms or changes in ion conductance, ultimately resulting in depolarization and action potentials in primary afferent nerve fibers. Various regions of the tongue are innervated by cranial nerves VII, IX, and X, which carry the signal to the medulla, from which it is transmitted to other regions of the brain, ultimately to the sensory cortex (Fig. 5.13).



Many people will recall seeing a diagram of the tongue, depicting localization of perception for sweet, salty, bitter, and sour tastes to various regions on its surface. Although it was thought for many years that different areas of the tongue are responsible for detection of various tastes, it is now believed that taste perception is not exclusively differentiated in this manner. In other words, the “taste map” seen in many texts is not well defined.